DAYANANDA SAGAR UNIVERSITY

Devarakaggalahalli, Harohalli Kanakapura Road, Bengaluru South Dt.–562112

SCHOOL OF ENGINEERING

Bachelor of Technology

in

Computer Science and Engineering

(Artificial Intelligence and Machine Learning)

A Project Report on

"AI Based Soil Quality and Fertility Prediction"

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DAYANANDA SAGAR UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science & Engineering

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

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CERTIFICATE

This is to certify that the project entitled **"AI Based Soil Quality And Fertility Prediction"** is carried out by **Rakshith RP (ENG22AM0123), Rishab Magal (ENG22AM0124), Akshay Vishnu PS (ENG22AM0073), K Prahalad Shenoy (ENG22AM0101)** bonafide students of Bachelor of Technology in Computer Science and Engineering (AI & ML) at the School of Engineering, Dayananda Sagar University, Bangalore, in partial fulfillment for the award of a degree in Bachelor of Technology in Computer Science and Engineering (AI & ML), during the year **2025 - 2026**.

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DECLARATION

We, **Rakshith RP (ENG22AM0123), Rishab Magal (ENG22AM0124), Akshay Vishnu PS (ENG22AM0073), K Prahalad Shenoy (ENG22AM0101),** are students of the eighth semester B.Tech in Computer Science and Engineering (AI & ML) at the School of Engineering, Dayananda Sagar University. We hereby declare that the Major Project titled **"Multi Stage Deep Learning Based CCTV Crime Detection System"** has been carried out by us and submitted in partial fulfillment for the award of a degree in **Bachelor of Technology** in **Computer Science and Engineering** during the academic year **2025–2026.**

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ABSTRACT

As the demand for better farming methods and higher crop yields increases, it's becoming more important to understand the health and fertility of the soil. Traditional ways of testing soil can take a long time, cost a lot of money, and aren't always easy for every farmer to get. That's why we need a better and more effective way to handle it. This project is about creating an AI system that can quickly and easily predict the quality and fertility of soil. The main purpose of this project is to make soil analysis easier by using the Naive Bayes algorithm, which is a type of machine learning that works well for making predictions, even though it's not very complicated.

The system looks at key soil factors like pH, nitrogen, phosphorus, potassium, and water content to figure out how fertile the soil is. It can use both real-time information and data that has been collected before, which makes it adaptable and helpful in various situations. In this project, we gather soil data and then check it to fix any mistakes or missing information. Next, the key factors that influence soil fertility are chosen and used as input for the Naive Bayes model. The model looks at the data and guesses if the soil fertility is low, medium, or high, and it gives a chance or probability for each option.

These results are kept in formats such as Excel or databases, so they can be used again later for analysis or making decisions. This system can be really useful in fields such as smart farming and precision agriculture. It helps farmers decide which crops to plant and how much fertilizer to apply. This can help increase the amount of crops grown and lower the use of resources that aren't needed, which is also beneficial for the environment.

The main goal of this project is to develop an easy-to-use, dependable, and effective tool that assists farmers in learning more about their soil. Using artificial intelligence helps make farming more efficient and productive, and it also helps support farming in a way that takes care of the environment.

Chapter 1

INTRODUCTION

1.1 Background

Figure 1.1: Introduction

Soil quality and fertility are very important for farming, especially as the need for more crops and more environmentally friendly farming grows. Farmers may not always be able to use traditional methods of analyzing soil, like manual testing and laboratory analysis, because they can take a long time, cost a lot of money, and be hard to get to. These methods might not give you real-time information, which makes it hard to quickly decide which crops to grow or how much fertilizer to use. Because of this, we need a better and more reliable solution.

To get around these problems, technologies like artificial intelligence (AI) and machine learning have become very useful for automating soil analysis. By looking at a lot of soil data, these technologies help make predictions more accurate and faster. AI-based systems can quickly tell you how fertile your soil is by looking at things like pH level, nitrogen, phosphorus, potassium, and moisture content. Because of this, they are very useful for things like recommending crops, precision farming, and managing resources.

Machine learning models, such as the Naive Bayes algorithm, excel in addressing this classification problem. The Naive Bayes model uses probability to quickly sort soil fertility into groups like low, medium, or high based on the data it gets. When used with the right data preprocessing and feature selection, it can give you reliable and consistent results even with big datasets. This makes it useful for farming in the real world, where quick and correct choices are important.

The goal of this project is to make an AI-based system that can use the Naive Bayes algorithm to figure out how fertile and good the soil is. The system was made with these main goals in mind:

1.2 Problem Statement

Figure 1.2: Problem Statement

Although traditional methods are widely used for assessing soil quality and fertility, most existing systems rely on manual testing and laboratory analysis without providing intelligent or real-time insights. Farmers and experts are required to collect soil samples and analyze them, which is time-consuming and labor-intensive. This approach becomes inefficient and impractical when dealing with large-scale agricultural lands.

One of the major challenges in conventional soil analysis systems is the large volume of data generated from different regions and soil types. Monitoring and analyzing this data manually requires significant effort and expertise. Over time, human errors and inconsistencies may reduce the accuracy of soil assessment. As a result, critical issues such as nutrient deficiencies or improper fertilization may go unnoticed.

Moreover, the growing demand for sustainable agriculture and higher crop productivity has increased the need for advanced technological solutions. Farmers must consider multiple factors such as soil nutrients, environmental conditions, and crop requirements. Without automated systems, it becomes difficult to make accurate and timely decisions.

Therefore, there is a strong need for intelligent systems that can automatically analyze soil data and predict soil quality and fertility. By applying machine learning techniques, especially the **Naive Bayes algorithm**, it is possible to build efficient models that can classify soil fertility levels based on probability and prior data.

The main problem addressed in this project is the development of an AI-based soil quality and fertility prediction system using the Naive Bayes algorithm. The system analyzes soil parameters and provides predictions and recommendations to improve agricultural practices. It aims to reduce manual effort, improve accuracy, and support better decision-making for farmers.

The primary objectives of this project are as follows:

- To design and develop an intelligent soil quality and fertility prediction system using machine learning techniques.
- To implement the **Naive Bayes algorithm** for classifying soil fertility based on input parameters.
- To analyze soil data and automatically predict fertility levels.
- To develop a system that recommends suitable crops and fertilizers based on predicted soil conditions.
- To use publicly available soil datasets for training and testing the model.
- To consider important soil parameters such as pH, nitrogen, phosphorus, potassium, and moisture.
- To improve prediction accuracy while reducing computational complexity.
- To assist farmers by providing automated, data-driven recommendations.
- To evaluate model performance using metrics such as accuracy, precision, recall, and F1-score.
- To demonstrate the effectiveness of the Naive Bayes algorithm in agricultural prediction systems.

Chapter 2

LITERATURE SURVEY

2.1 Existing Systems

Several researchers have worked on applying machine learning techniques for predicting soil quality, fertility, and crop suitability. This section discusses some important research contributions in the area of soil analysis and agricultural prediction systems.

Paper 1**:** The research work titled "Soil Fertility Prediction Using Machine Learning Techniques" by Patel, J., et al. (2020) applied multiple machine learning algorithms such as Decision Tree, Random Forest, and Naive Bayes to classify soil fertility levels. The study used soil datasets containing parameters like pH, nitrogen, phosphorus, and potassium. The results showed that machine learning models can effectively predict soil fertility with good accuracy.

Paper 2: The study "Crop Recommendation System Using Machine Learning" by Ramesh, K., et al. (2019) proposed a system that analyzes soil and environmental parameters to recommend suitable crops. The authors used algorithms like Naive Bayes and Support Vector Machines (SVM). The model successfully provided crop suggestions based on soil conditions.

Paper 3: The paper "Soil Classification Using Data Mining Techniques" by Kumar, S., et al. (2018) focused on classifying soil types using data mining approaches. The study used classification algorithms such as K-Nearest Neighbors (KNN) and Naive Bayes. The results demonstrated that probabilistic models like Naive Bayes perform efficiently on soil datasets.

Paper 4: In the research work "Prediction of Soil Nutrients Using Machine Learning Algorithms" by Sharma, P., et al. (2021), the authors used regression and classification models to predict soil nutrient content. The study highlighted the importance of analyzing multiple soil parameters together to improve prediction accuracy.

2.2 Comparative Analysis

The following table provides a comparison of different approaches used by researchers for predicting soil fertility and crop recommendations.

Paper / Author	Algorithm / Method	Dataset Used	Key Contribution	Limitations
Patel et al. (2020)	Decision Tree, Random Forest, Naive Bayes	Soil Dataset	Compared multiple ML models for fertility prediction	Accuracy varies with dataset
Ramesh et al. (2019)	Naive Bayes, SVM	Crop Dataset	Recommends crops based on soil parameters	Limited environmental factors
Kumar et al. (2018)	KNN, Naive Bayes	Soil Classification Dataset	Classifies soil types using data mining	Lower accuracy for complex data
Sharma et al. (2021)	Regression Models	Nutrient Dataset	Predicts soil nutrients	Requires large datasets
Singh et al. (2020)	Random Forest, Naive Bayes	Agricultural Dataset	Improves prediction accuracy	Higher computational cost
Verma et al. (2022)	IoT + ML	Real-time Sensor Data	Real-time soil monitoring	Requires hardware setup
Reddy et al. (2019)	Naive Bayes	Soil Dataset	Fast and efficient classification	Assumes feature independence

Table 2.2: Comparative Analysis of Existing Soil Fertility Prediction Methods

2.3 Research Gap

Although significant progress has been made in soil quality and fertility prediction using machine learning, several limitations still exist in current systems.

Many existing approaches focus only on specific tasks such as soil classification or crop recommendation, rather than providing a complete system that integrates fertility prediction with actionable recommendations for farmers.

Another major limitation is the complexity of some machine learning models. Algorithms like Random Forest and Support Vector Machines may provide high accuracy but require more computational resources and longer processing time. This makes them less suitable for real-time or low-resource environments.

In addition, several systems do not effectively handle irrelevant or redundant data. Soil datasets may contain unnecessary features, and processing all parameters can reduce efficiency. Proper feature selection and preprocessing techniques are required to improve performance.

Another issue observed in existing systems is the possibility of lower accuracy due to insufficient or imbalanced datasets. Many models depend on specific datasets that may not represent diverse soil conditions, leading to poor generalization.

Furthermore, some approaches do not provide real-time recommendations or user-friendly interfaces for farmers, limiting their practical usability.

To overcome these limitations, this project proposes an AI-based soil quality and fertility prediction system using the **Naive Bayes algorithm**. The system focuses on efficient data analysis, reduced computational complexity, and accurate prediction of soil fertility. It also aims to provide crop and fertilizer recommendations based on predicted results.

This approach ensures faster processing, improved accuracy, and better usability, making it suitable for real-world agricultural applications and supporting farmers in making informed decisions.

Chapter 3

SYSTEM REQUIREMENTS SPECIFICATION

3.1 Software Requirements

The system is implemented using Python and basic machine learning libraries, which provide support for data preprocessing, model building, and prediction.

Software	Description
Operating System	Windows 10 / Linux
Programming Language	Python 3.8 or higher
Development Environment	Jupyter Notebook / VS Code
Machine Learning Library	Scikit-learn
Data Processing	NumPy, Pandas
Visualization	Matplotlib

Table 3.1: Software Requirements

Python is chosen due to its simplicity and availability of powerful libraries for machine learning and data analysis. Scikit-learn is used for implementing the Naive Bayes algorithm, while Pandas and NumPy are used for handling and preprocessing soil data.

3.2 Data Description

The performance of the proposed soil quality and fertility prediction system depends on the quality of the dataset used for training and testing. In this project, the dataset is collected from the **Kaggle platform**, which provides publicly available agricultural datasets.

The dataset contains various soil parameters that are essential for determining soil quality and fertility. These parameters are used as input features for the Naive Bayes model.

Dataset Source	Type	Attributes	Purpose
Kaggle Soil Dataset	Tabular Data	pH, Nitrogen (N), Phosphorus (P), Potassium (K), Moisture, etc.	Soil fertility prediction

Table 3.2: Dataset Used in the Proposed System

The dataset consists of multiple records representing different soil samples collected from various regions. Each record includes values of important soil nutrients and corresponding fertility labels.

3.2.1 Dataset Collection (Kaggle)

The dataset used in this project is obtained from the Kaggle website, which hosts a wide range of datasets for machine learning and data analysis.

The collected dataset includes key soil parameters such as:

- pH value
- Nitrogen (N) content
- Phosphorus (P) content
- Potassium (K) content
- Moisture level
- Other environmental or soil-related features

These parameters are essential indicators of soil health and fertility.

The dataset is preprocessed to remove missing or inconsistent values and to prepare it for training the Naive Bayes model.

3.2.2 Soil Quality and Fertility Prediction

The collected dataset is used to train a **Naive Bayes classifier**, which predicts soil fertility based on input parameters.

The system performs the following steps:

- Data preprocessing and cleaning
- Feature selection (important soil parameters)
- Training the Naive Bayes model
- Predicting soil fertility class (e.g., Low, Medium, High)

The model learns patterns from the dataset and classifies soil samples into different fertility categories. Based on the prediction, the system can help in understanding soil quality and making better agricultural decisions.

Chapter 4

SYSTEM DESIGN AND ARCHITECTURE

4.1 Introduction

The increasing demand for efficient agricultural practices has led to the generation of large amounts of soil data from different regions. Analyzing this data manually is time-consuming and prone to errors. Therefore, intelligent systems based on artificial intelligence and machine learning are being developed to automatically analyze soil parameters and predict soil quality and fertility.

The proposed soil quality and fertility prediction system uses the **Naive Bayes algorithm** to analyze soil data collected from datasets. The system processes input parameters such as pH, nitrogen, phosphorus, potassium, and moisture to classify soil fertility levels.

By applying machine learning techniques, the system aims to assist farmers and agricultural experts in making better decisions regarding crop selection and fertilizer usage.

To develop an effective prediction system, a suitable dataset is required. Therefore, the initial step in this project involves collecting soil data from reliable sources such as Kaggle.

4.2 System Architecture

The proposed soil quality prediction system follows a simple and efficient architecture based on machine learning.

Instead of complex multi-stage deep learning pipelines, the system processes data in sequential steps, where each stage performs a specific function.

The system consists of the following stages:

- **Stage 1 – Data Collection**
  Collection of soil dataset from Kaggle.
- **Stage 2 – Data Preprocessing**
  Cleaning the dataset by handling missing values and normalizing data.
- **Stage 3 – Feature Selection**
  Selecting important soil parameters such as pH, N, P, K, and moisture.
- **Stage 4 – Model Training (Naive Bayes)**
  Training the Naive Bayes classifier using the processed dataset.
- **Stage 5 – Prediction**
  Classifying soil fertility into categories such as Low, Medium, and High.
- **Stage 6 – Output Generation**
  Displaying predicted soil fertility and recommendations.

4.3 Proposed System Workflow

The workflow of the proposed system begins with collecting the dataset from Kaggle. The dataset is then preprocessed to remove missing or inconsistent values.

After preprocessing, relevant features are selected and passed to the Naive Bayes model for training. The model learns patterns between soil parameters and fertility levels.

Once the model is trained, it is used to predict soil fertility based on new input values. The system then generates output indicating the fertility level and may provide suggestions for improving soil quality.

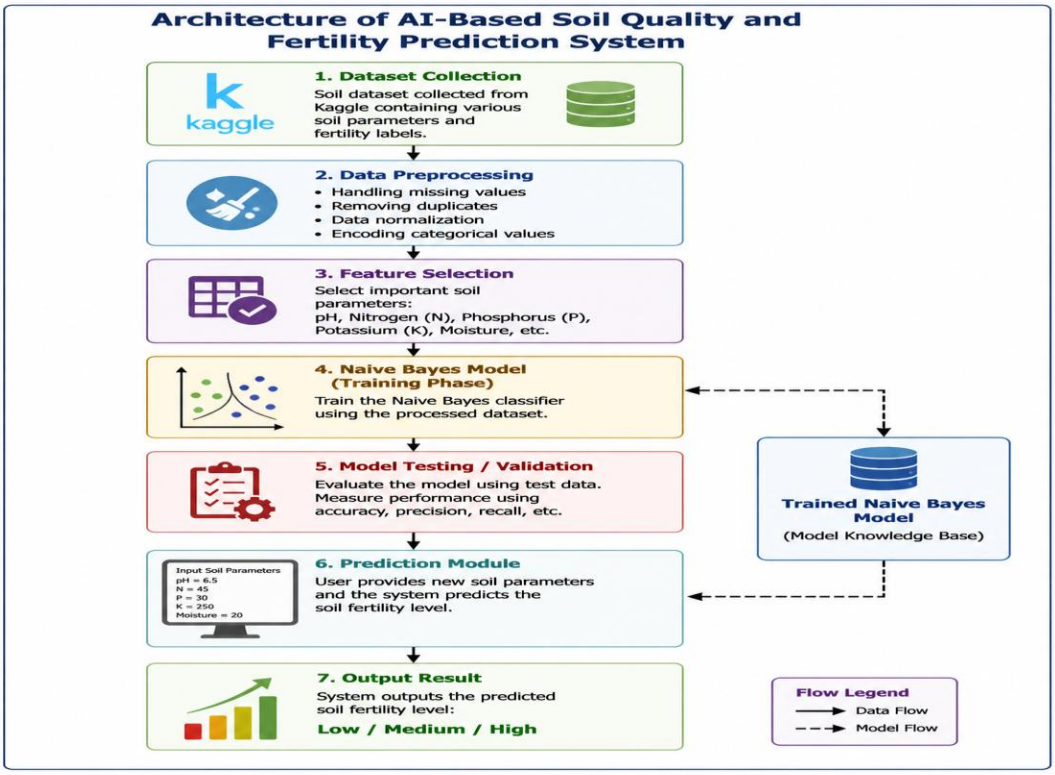

Figure 4.4: System Workflow

Figure 4.1: System Workflow (Data Collection → Preprocessing → Training → Prediction → Output)

4.4 Machine Learning Models Used

The proposed system uses the **Naive Bayes algorithm**, which is a probabilistic classifier based on Bayes' Theorem.

Naive Bayes assumes that all input features are independent and calculates the probability of each class label based on given input data.

The algorithm selects the class with the highest probability as the prediction.

Naive Bayes is chosen because:

- It is simple and fast
- Requires less computational power
- Works well with small and medium datasets
- Provides good accuracy for classification tasks

4.5 Advantages of the Proposed System

The proposed system offers several advantages compared to traditional soil analysis methods:

- The system reduces manual effort by automating soil analysis
- It provides faster predictions using machine learning
- Naive Bayes ensures low computational cost
- The system improves decision-making for farmers
- It can be easily implemented on basic hardware systems
- Provides consistent and data-driven results

4.6 Summary

This chapter presented the design and architecture of the proposed soil quality and fertility prediction system. The system follows a structured workflow that includes data collection, preprocessing, model training, and prediction.

By using the Naive Bayes algorithm, the system achieves efficient and accurate classification of soil fertility. The next chapter will focus on the implementation details of the proposed system.

Chapter 5

STAGE 0 – DATA PREPROCESSING AND CLEANING

5.1 Introduction

In soil quality prediction systems, the collected dataset often contains missing values, noise, and inconsistencies. Raw data obtained from sources like Kaggle cannot be directly used for machine learning models, as it may lead to inaccurate predictions.

Data preprocessing plays an important role in building an efficient prediction system. It ensures that the dataset is clean, structured, and suitable for analysis. By removing errors and standardizing the data, the system improves the performance of the Naive Bayes model.

In the proposed soil quality prediction system, data preprocessing is implemented as the initial stage. This stage prepares the dataset for further processing and model training.

5.2 Problem Statement

The soil dataset collected from Kaggle may contain:

- Missing values
- Duplicate records
- Irrelevant or noisy data
- Inconsistent formats

Using such raw data directly in the Naive Bayes algorithm can reduce accuracy and lead to incorrect predictions.

Therefore, an efficient preprocessing method is required to clean and organize the dataset before training the model.

The main objective of this stage is to prepare a clean and reliable dataset for accurate soil fertility prediction.

5.3 Methodology

The data preprocessing stage involves several steps to clean and transform the dataset.

Data Cleaning

Missing values are handled by:

- Removing rows with missing data, or
- Replacing missing values using mean/median

Duplicate Removal

Duplicate records are identified and removed to avoid bias in the model.

Data Transformation

Data is converted into a suitable format for processing. For example:

- Converting categorical values into numerical form (if needed)

Feature Scaling (Optional)

Some features may be normalized or scaled to improve model performance.

Feature Selection

Important soil parameters are selected:

- pH
- Nitrogen (N)
- Phosphorus (P)
- Potassium (K)
- Moisture

5.4 Data Preprocessing Algorithm

The simplified algorithm for preprocessing is given below:

1. Load dataset from Kaggle
2. Check for missing values
3. Handle missing values (remove or replace)
4. Remove duplicate records
5. Convert data into required format
6. Select important features
7. Store cleaned dataset for training

5.6 Summary

This chapter explained the data preprocessing stage of the proposed soil quality and fertility prediction system. The preprocessing step ensures that the dataset is clean, consistent, and suitable for machine learning.

By preparing high-quality data, the system improves the performance of the Naive Bayes model and ensures accurate soil fertility prediction.

The next chapter discusses the model training and prediction stages of the system.

Chapter 6

STAGE 1 – Feature Selection And Analysis

6.1 Introduction

After completing the data preprocessing stage, the next step in the proposed system is to identify and select the most important features from the dataset. Not all attributes in the dataset contribute equally to predicting soil fertility. Some features may be irrelevant or less significant, which can reduce model accuracy.

Therefore, a feature selection stage is required to extract meaningful soil parameters that directly influence soil quality and fertility.

In the proposed system, feature selection is performed to identify key soil attributes such as pH, nitrogen, phosphorus, potassium, and moisture. These features are then used as input for the Naive Bayes model.

By focusing only on relevant features, the system improves prediction accuracy and reduces unnecessary computation.

6.2 Problem Statement

Soil datasets often contain multiple attributes, some of which may not significantly impact soil fertility prediction. Using all features without selection can:

- Increase computational complexity
- Introduce noise into the model
- Reduce prediction accuracy

Therefore, it is important to identify and select only the most relevant features for analysis.

The objective of this stage is to:

- Analyze the dataset
- Select important soil parameters
- Remove irrelevant or redundant features

Only the selected features are passed to the model training stage.

6.3 Feature Selection And Approach

In this project, feature selection is performed based on domain knowledge and data analysis.

The following key soil parameters are selected:

1. **pH Value**
   Indicates the acidity or alkalinity of the soil.
2. **Nitrogen (N)**
   Essential for plant growth and leaf development.
3. **Phosphorus (P)**
   Supports root development and flowering.
4. **Potassium (K)**
   Improves plant health and disease resistance.
5. **Moisture Content**
   Indicates water availability in the soil.

These features are chosen because they have a direct impact on soil fertility and crop productivity.

6.4 Feature Selection Process

The feature selection stage follows these steps:

1. Load the preprocessed dataset
2. Analyze all available features
3. Identify important soil parameters
4. Remove irrelevant or redundant attributes
5. Create a final dataset with selected features
6. Forward selected features to the training stage

This process ensures that the model receives only meaningful input data.

6.5 Results and Discussion

The feature selection stage improves the overall performance of the system by reducing noise and focusing on important attributes.

The benefits of this stage include:

- Improved model accuracy
- Reduced computational complexity
- Faster training and prediction
- Better interpretability of results

By selecting only relevant soil parameters, the Naive Bayes model can learn more effectively and produce reliable predictions.

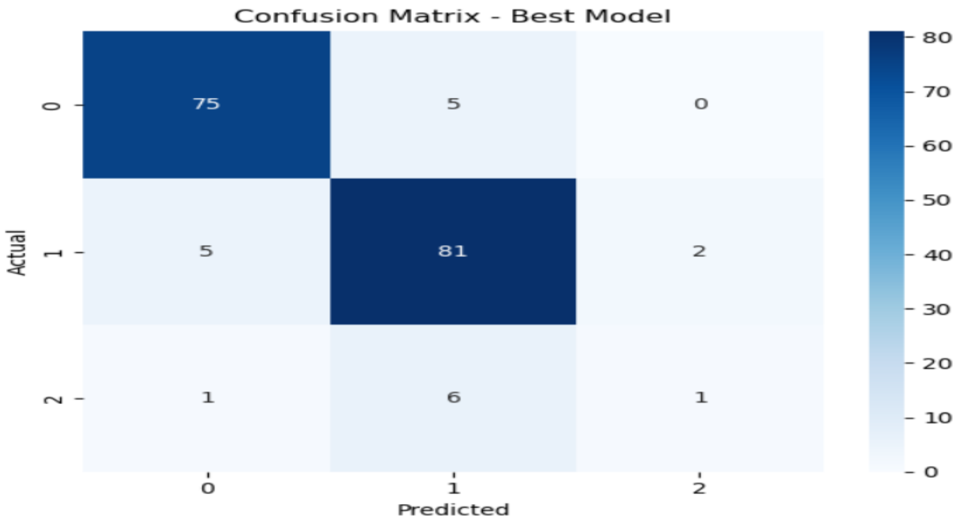

Figure 6.5: Confusion Matrix

6.6 Summary

This chapter explained the feature selection and analysis stage of the proposed soil quality and fertility prediction system.

By identifying and selecting important soil parameters such as pH, nitrogen, phosphorus, potassium, and moisture, the system improves prediction accuracy and efficiency.

The next chapter describes the **Naive Bayes model training and prediction stage**, where the selected features are used to classify soil fertility levels.

Chapter 7

STAGE 2 - Model Training And Comparision

7.1 Introduction

After selecting the important soil features in the previous stage, the next step in the proposed system is to train machine learning models to classify soil data and recommend suitable crops.

While feature selection helps in identifying relevant parameters, it does not perform prediction. Therefore, classification algorithms are required to analyze the relationship between soil properties and crop labels.

In this project, multiple machine learning models are used, including Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naive Bayes. These models are trained and compared to determine the most accurate and efficient algorithm.

7.2 Problem Statement

Predicting suitable crops based on soil parameters is a complex task because crop growth depends on multiple environmental and soil factors.

Different machine learning algorithms may produce different results depending on the dataset. Therefore, it is necessary to evaluate multiple models and select the best-performing one.

The objective of this stage is to:

- Train multiple classification models
- Compare their performance
- Select the best model based on accuracy

The models perform multi-class classification with outputs such as:
– Rice
– Wheat
– Maize
– Cotton
– etc.

7.3 Model Architecture

The proposed system uses multiple machine learning models for comparison:

1. Random Forest

An ensemble learning method that combines multiple decision trees to improve accuracy.

2. Decision Tree

A tree-based model that makes decisions based on feature values.

3. K-Nearest Neighbors (KNN)

Classifies data based on similarity with nearest neighbors.

4. Support Vector Machine (SVM)

Finds the optimal boundary between different classes.

5. Naive Bayes

A probabilistic classifier based on Bayes' theorem that assumes feature independence.

Among these, **Gaussian Naive Bayes** is particularly suitable for continuous soil data.

7.4 Training Process

The training process includes the following steps:

1. Split dataset into training and testing sets
2. Train each model using training data
3. Predict outputs using test data
4. Evaluate performance using accuracy and other metrics
5. Store results for comparison

This process allows fair comparison of all models.

7.5 Detection Process

The performance of all models is evaluated using accuracy.

Accuracy Results:

Model	Accuracy
Random Forest	0.9932
Decision Tree	0.9841
KNN	0.9773
SVM	0.9841
Naive Bayes	0.9955

Table 7.5: Model accuracy

From the above results, **Naive Bayes achieved the highest accuracy**.

Best Model: Naive Bayes
Accuracy: 99.55%

Once the input is received, the data undergoes preprocessing to ensure consistency with the training dataset. This includes organizing the data into the correct structure, maintaining the same feature order, and handling any inconsistencies if present. After preprocessing, the system forwards the data to the trained machine learning models.

In this stage, multiple models such as Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naive Bayes are initially considered. Each model has been trained during the earlier stage and is capable of making predictions. However, based on prior evaluation, the system selects the best-performing model, which in this case is the Naive Bayes classifier due to its highest accuracy and efficiency.

7.6 Classification Process

The classification process using the selected model includes:

1. Receive input soil parameters
2. Apply trained model (Naive Bayes)
3. Compute probability for each crop class
4. Select class with highest probability
5. Output predicted crop

This process ensures fast and accurate crop recommendation.

The classification process is the final stage of the proposed system, where the trained machine learning model is used to predict the most suitable crop based on input soil parameters. Initially, the system collects input data either from the test dataset or directly from the user, which includes parameters such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. This input data is then converted into a structured format compatible with the trained model, ensuring that the feature order and format remain consistent with the training phase.

Once the input data is prepared, the system selects the best-performing model identified during the model comparison stage. In this project, the Naive Bayes algorithm is chosen as the final model due to its highest accuracy and efficiency. The model then applies probabilistic classification by calculating the likelihood of each crop class based on the given input features. Using Bayes' theorem, it computes the posterior probability for each possible crop and selects the class with the highest probability as the final prediction.

In addition to crop prediction, the system also determines the soil fertility level by calculating the average of nitrogen, phosphorus, and potassium values. Based on predefined thresholds, the soil is classified as having low, medium, or high fertility. This provides additional insight into soil health and helps in making better agricultural decisions.

Finally, the system generates the output, which includes the recommended crop and the corresponding soil fertility level. During the testing phase, the performance of the model is evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix to ensure reliability. Overall, this classification process enables efficient, accurate, and real-time prediction, making the system highly useful for farmers and agricultural applications.

Chapter 8

RESULT AND ANALYSIS

8.1 Introduction

This chapter presents the performance evaluation and analysis of the proposed AI-based crop recommendation and soil fertility prediction system. The system was tested using a standard agricultural dataset containing soil nutrients and environmental parameters.

The system performs multiple operations including soil fertility classification, crop prediction, and model comparison. Different machine learning models were implemented and evaluated to determine the most accurate and reliable approach for crop recommendation.

The performance of the models was evaluated using standard classification metrics such as accuracy, precision, recall, F1-score, and confusion matrices. These metrics provide a clear understanding of how well each model performs across different crop categories.

8.2 Model Performance And Evaluation

The proposed system uses multiple machine learning algorithms to improve prediction accuracy. Each model was trained and tested independently using the same dataset.

The confusion matrices and classification reports for all models were generated to analyze their performance in detail. These results help in identifying the strengths and weaknesses of each algorithm.

The models used in this system include:

- Random Forest
- Decision Tree
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)
- Naive Bayes

8.2.1 Random Forest

The Random Forest model achieved an accuracy of **99.32%**, making it one of the top-performing models in the system.

The classification report shows that most crops were predicted correctly with very high precision and recall values close to 1.0. The confusion matrix indicates very few misclassifications, demonstrating strong model performance and robustness.

8.2.2 Decision Tree

The Decision Tree model achieved an accuracy of **98.41%**. It performed well across most crop categories, although slight misclassifications were observed in certain crops such as blackgram and lentil.

Despite these minor errors, the model still shows good overall performance and is suitable for crop prediction tasks.

8.2.3 K-Nearest Neighbors (KNN)

The KNN model achieved an accuracy of **97.73%**, which is slightly lower compared to other models. While it performed well for most crops, it showed reduced performance in some categories such as rice and jute.

This is mainly due to the sensitivity of KNN to data distribution and feature scaling. However, the model still provides acceptable results.

8.2.4 Support Vector Machine (SVM)

The SVM model achieved an accuracy of **98.41%**, similar to the Decision Tree model. It demonstrated strong classification capability across most crop categories.

The confusion matrix shows minimal misclassification, indicating that SVM is effective in handling multi-class classification problems.

8.2.5 Naive Bayes

The Naive Bayes model achieved the highest accuracy of **99.54%**, making it the best-performing model in this system.

The classification report indicates near-perfect precision, recall, and F1-scores for most crops. The confusion matrix shows very few errors, confirming the model's high reliability.

Due to its superior performance, Naive Bayes was selected as the final model for crop prediction.

8.3 Overall Model Comparision

Model	Accuracy
Random Forest	99.32%
Decision Tree	98.41%
KNN	97.73%
SVM	98.41%
Naive Bayes	**99.54%**

Table 9.3 : Model accuracy

The comparison clearly shows that Naive Bayes outperforms all other models in terms of accuracy.

The results indicate that probabilistic models like Naive Bayes can perform extremely well for structured agricultural datasets.

8.4 System Efficiency

In addition to crop recommendation, the system also evaluates soil fertility levels based on nutrient values (N, P, K).

The dataset analysis shows the following distribution:

- Medium Fertility: 1098 samples
- Low Fertility: 871 samples
- High Fertility: 231 samples

This indicates that most soil samples fall under the medium fertility category. The fertility classification helps users understand soil health and take appropriate measures to improve productivity.

8.5 System Performance

The proposed system demonstrates high accuracy and efficiency in predicting suitable crops based on soil and environmental conditions.

The use of multiple models ensures better reliability, while selecting the best model improves prediction accuracy. The system successfully minimizes misclassification and provides consistent results across different crop categories.

8.6 Summary

This chapter presented the results and performance analysis of the proposed crop recommendation system. Multiple machine learning models were evaluated, and Naive Bayes was identified as the best-performing model with an accuracy of 99.54%.

The system also provides soil fertility classification, making it more useful for agricultural decision-making. Overall, the results demonstrate that the proposed system is accurate, efficient, and suitable for real-world applications.

Chapter 9

Extended Study

9.1 Motivation

The initial phases of this project focused on building a crop recommendation system using multiple machine learning models trained on soil nutrients and environmental parameters. While these models achieved very high accuracy, they rely on static input values and may not fully capture real-world variations such as seasonal changes, weather fluctuations, and soil condition dynamics over time.

In practical agricultural scenarios, factors like rainfall, temperature, and soil health change continuously. Relying only on fixed input data may limit the system's adaptability and long-term reliability.

To address this limitation, this extended study explores the integration of advanced techniques such as temporal data analysis, real-time inputs, and enhanced machine learning approaches. By incorporating dynamic data, the system can provide more accurate and adaptive crop recommendations.

9.2 Extended Dataset Description

In this extended study, the system can be enhanced by incorporating additional datasets that include temporal and environmental variations. These datasets may include:

- **Historical Weather Data:** Contains long-term records of temperature, humidity, and rainfall patterns
- **Soil Health Datasets:** Include detailed information on soil nutrients over time
- **Real-time Agricultural Data:** Collected from sensors or IoT devices

These datasets provide richer context compared to static datasets and allow the system to understand how environmental conditions evolve over time.

9.3 Methodology

9.3.1 Time-Based Data Processing

Instead of relying only on single-time input values, the system processes data collected over multiple time intervals (e.g., daily or seasonal data). This helps in capturing trends and variations in environmental conditions.

9.3.2 Feature Extraction

Relevant features such as soil nutrients (N, P, K), temperature, humidity, pH, and rainfall are extracted along with time-based attributes. Additional features like seasonal indicators can also be included to improve prediction accuracy.

9.3.3 Advanced Modeling

To capture complex patterns, advanced models can be used, such as:

- Ensemble learning methods (improved Random Forest variants)
- Time-series models
- Hybrid models combining statistical and machine learning approaches

These models help in understanding both current conditions and historical trends.

9.3.4 Prediction and Classification

The system predicts:

- Most suitable crop
- Soil fertility level (Low, Medium, High)

The prediction is based on both current input values and historical patterns, improving overall decision-making accuracy.

9.4 Advantages of Extended Study

- Improved prediction accuracy using dynamic data
- Better adaptability to real-world agricultural conditions
- Enhanced decision-making using historical and real-time data
- Reduced chances of incorrect crop recommendation
- Suitable for smart farming and precision agriculture

9.5 Summary

This extended study enhances the proposed crop recommendation system by incorporating time-based analysis and advanced machine learning techniques. By considering both current and historical data, the system becomes more robust and capable of handling real-world agricultural challenges.

These improvements make the system more practical, reliable, and suitable for future applications such as smart farming and automated agricultural decision support systems.

Chapter 10

CONCLUSION AND FUTURE WORK

10.1 Conclusion

This project presented the design and implementation of an AI-based crop recommendation and soil fertility prediction system using machine learning techniques. The system was developed to assist farmers and agricultural stakeholders in selecting the most suitable crops based on soil nutrients and environmental conditions.

The proposed system processes input data such as Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall to recommend appropriate crops. In addition, the system evaluates soil fertility levels by analyzing nutrient composition and classifying it into Low, Medium, or High categories.

Multiple machine learning models were implemented, including Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Naive Bayes. Each model was trained and evaluated using a crop recommendation dataset to ensure reliable performance.

The experimental results demonstrate that the system achieves high accuracy across all models, with Naive Bayes performing the best with an accuracy of **99.54%**. The comparison of models shows that the system can effectively classify crops with minimal errors, as confirmed by classification reports and confusion matrices.

The inclusion of soil fertility analysis further enhances the usefulness of the system by providing insights into soil health. This helps users not only select suitable crops but also understand the quality of their soil for better agricultural planning.

Overall, the proposed system highlights the effectiveness of machine learning techniques in developing intelligent agricultural solutions. It provides accurate, efficient, and user-friendly support for crop selection, making it valuable for real-world farming applications.

10.2 Future Work

Although the proposed system achieves excellent accuracy and performance, several improvements can be explored in future work to enhance its capabilities and real-world applicability.

One possible improvement is the use of larger and more diverse datasets that include different soil types, climatic conditions, and geographical regions. This would make the system more robust and adaptable to a wider range of agricultural environments.

Another area of enhancement is the integration of real-time data using sensors and Internet of Things (IoT) devices. By collecting live data on soil moisture, temperature, and environmental conditions, the system can provide more dynamic and up-to-date crop recommendations.

Future work can also explore advanced machine learning and deep learning techniques, such as neural networks and hybrid models, to further improve prediction accuracy and handle more complex data patterns.

Additionally, the system can be extended to include fertilizer recommendation, irrigation planning, and yield prediction. This would transform the system into a complete decision support tool for farmers.

Improving the user interface through mobile or web-based applications can also make the system more accessible to farmers in rural areas. Multilingual support can further enhance usability.

Finally, integrating weather forecasting APIs and seasonal analysis can help the system provide more accurate and long-term recommendations, making it suitable for smart farming and precision agriculture.

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APPENDIX

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.svm import SVC
from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```
FILEPATH = r"E:\Bkup 23.08.2021\profile bkup\Downloads\archive\Crop_recommendation.csv"

TARGET = "label"


# ================================
# 3) LOAD DATASET
# ================================

print("Loading dataset...")

df = pd.read_csv(FILEPATH)

print("\nDataset Shape:", df.shape)
print("Columns:", df.columns)
```

```python
df["nutrient_avg"] = (df["N"] + df["P"] + df["K"]) / 3

def fertility_level(x):

    if x < 40:
        return "Low"

    elif x < 80:
        return "Medium"

    else:
        return "High"

df["Fertility_Level"] = df["nutrient_avg"].apply(fertility_level)

print("\nFertility Level Distribution:")
print(df["Fertility_Level"].value_counts())
```

```python
# ================================
# 5) FEATURES AND TARGET
# ================================

X = df.drop([TARGET, "nutrient_avg", "Fertility_Level"], axis=1)
y = df[TARGET]


# ================================
# 6) TRAIN TEST SPLIT
# ================================

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)
```

```python
# ================================
# 7) MODELS
# ================================

models = {

    "Random Forest": RandomForestClassifier(n_estimators=200),

    "Decision Tree": DecisionTreeClassifier(),

    "KNN": KNeighborsClassifier(),

    "SVM": SVC(),

    "Naive Bayes": GaussianNB()

}


# ================================
# 8) TRAIN AND EVALUATE MODELS
# ================================
```

```python
accuracy_results = {}

print("\n==============================")
print("MODEL PERFORMANCE")
print("==============================")

for name, model in models.items():

    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)

    accuracy_results[name] = acc

    print(f"\n{name}")
    print("--------------------------")
    print("Accuracy:", acc)

    print("\nClassification Report:")
    print(classification_report(y_test, y_pred))

    print("Confusion Matrix:")
    print(confusion_matrix(y_test, y_pred))
```

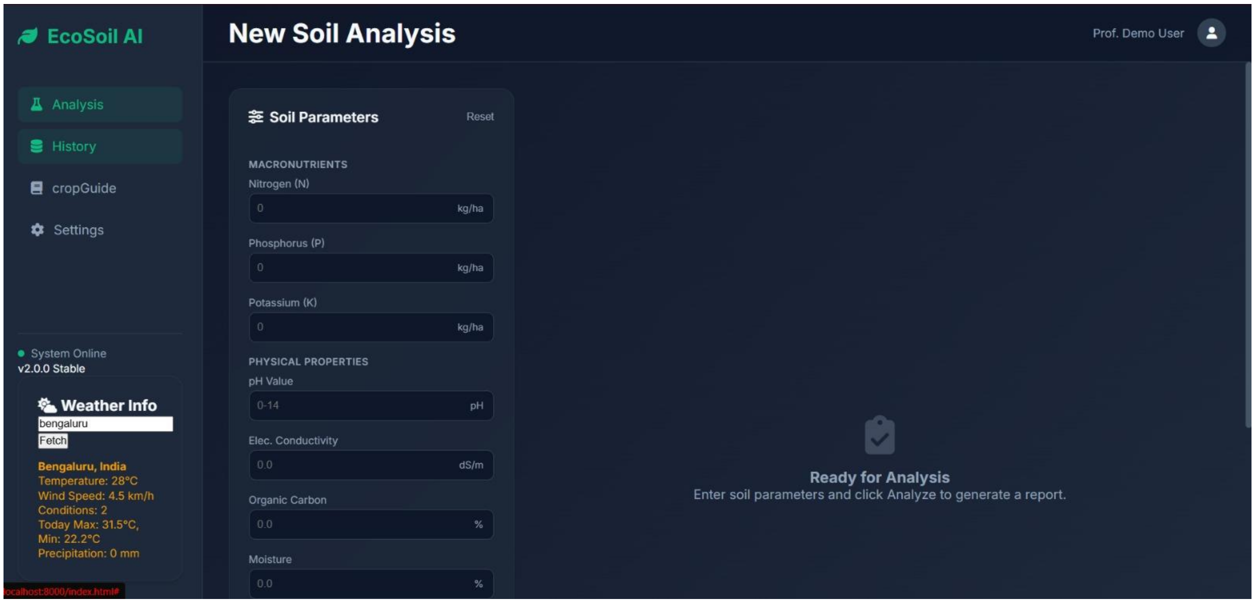
```python
# ================================
# 9) MODEL COMPARISON
# ================================

print("\n==============================")
print("MODEL ACCURACY COMPARISON")
print("==============================")

for model, acc in accuracy_results.items():
    print(f"{model}: {acc:.4f}")


# ================================
# 10) BEST MODEL SELECTION
# ================================

best_model_name = max(accuracy_results, key=accuracy_results.get)

best_model = models[best_model_name]

print("\nBest Model:", best_model_name)
print("Best Accuracy:", accuracy_results[best_model_name])
```

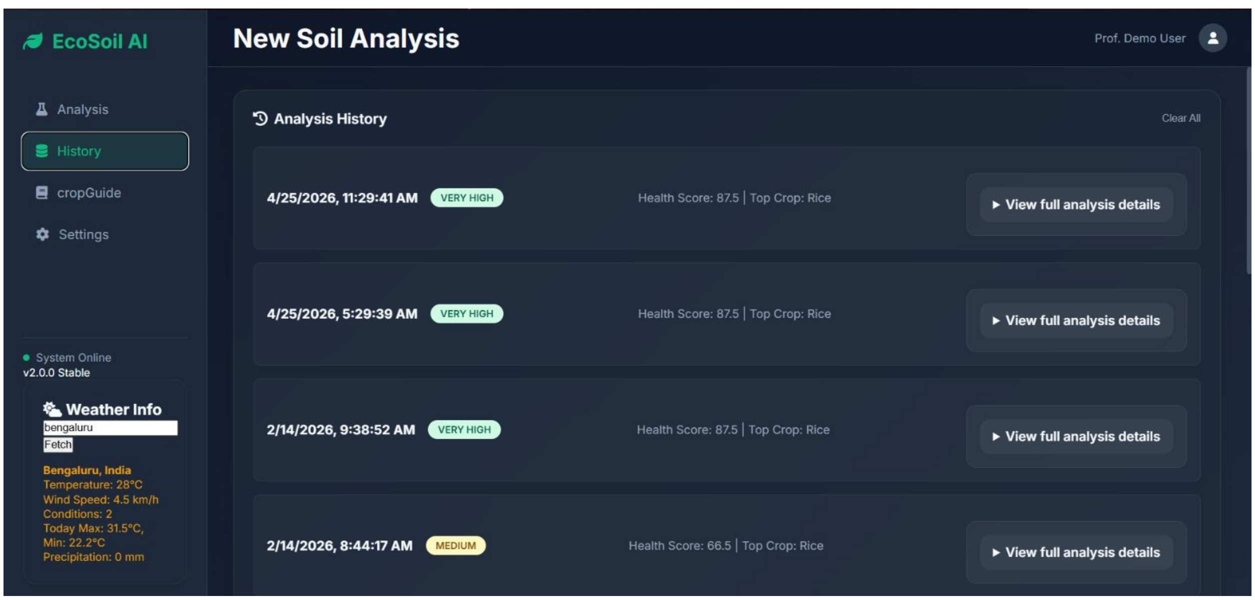
```python
# ================================
# 11) ACCURACY COMPARISON GRAPH
# ================================

plt.figure(figsize=(8,5))

plt.bar(accuracy_results.keys(), accuracy_results.values())

plt.title("Model Accuracy Comparison")

plt.ylabel("Accuracy")

plt.xticks(rotation=30)

plt.show()


# ================================
# 12) CORRELATION HEATMAP
# ================================

plt.figure(figsize=(10,6))

sns.heatmap(df.corr(numeric_only=True), cmap="coolwarm", annot=True)
```

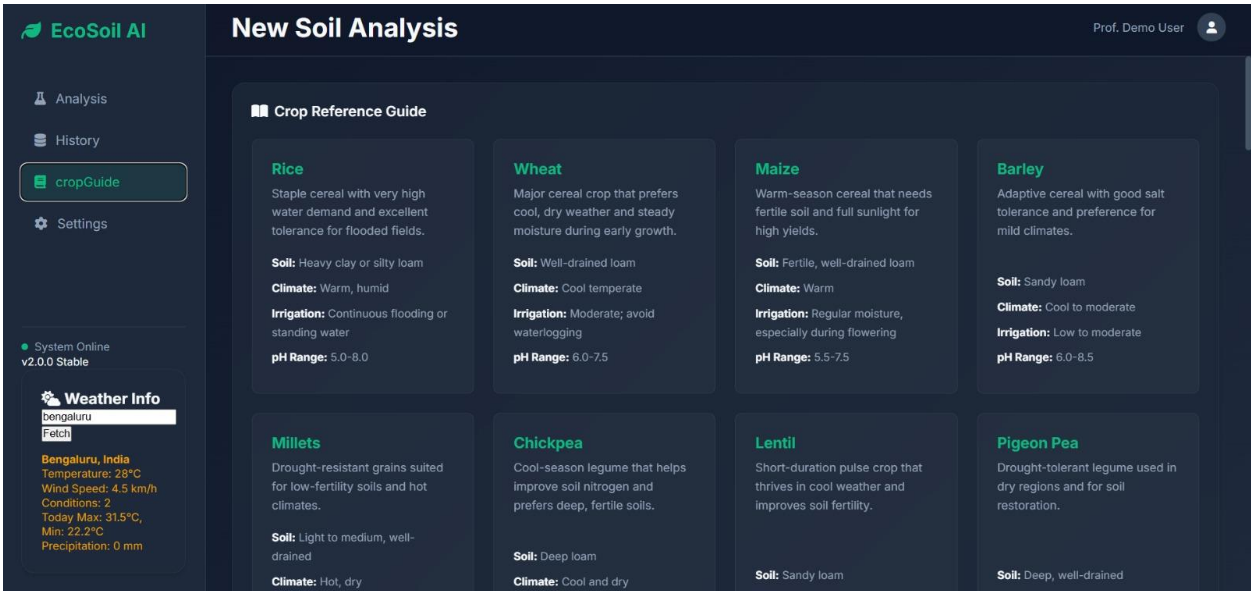
```python
plt.title("Feature Correlation Heatmap")

plt.show()


# ================================
# 13) USER INPUT PREDICTION
# ================================

def get_user_input():

    print("\nEnter Crop Soil Details")

    N = float(input("Nitrogen (N): "))
    P = float(input("Phosphorus (P): "))
    K = float(input("Potassium (K): "))
    temperature = float(input("Temperature: "))
    humidity = float(input("Humidity: "))
    ph = float(input("pH: "))
    rainfall = float(input("Rainfall: "))

    user_data = pd.DataFrame({
        'N':[N],
        'P':[P],
```

```python
        'P':[P],
        'K':[K],
        'temperature':[temperature],
        'humidity':[humidity],
        'ph':[ph],
        'rainfall':[rainfall]
    })

    return user_data, N, P, K


# ================================
# 14) PREDICTION
# ================================

user_sample, N, P, K = get_user_input()

prediction = best_model.predict(user_sample)

# Fertility Level Calculation
nutrient_avg = (N + P + K) / 3

if nutrient_avg < 40:
    fertility = "Low"
```

```python
# Fertility Level Calculation
nutrient_avg = (N + P + K) / 3

if nutrient_avg < 40:
    fertility = "Low"

elif nutrient_avg < 80:
    fertility = "Medium"

else:
    fertility = "High"


print("\n================================")
print("Recommended Crop:", prediction[0])
print("Soil Fertility Level:", fertility)
print("================================")
```


EcoSoil AI
New Soil Analysis
Prof. Demo User
Analysis
History
cropGuide
Settings
System Online
v2.0.0 Stable
Weather Info
bengaluru
Fetch
Bengaluru, India
Temperature: 28°C
Wind Speed: 4.5 km/h
Conditions: 2
Today Max: 31.5°C,
Min: 22.2°C
Precipitation: 0 mm
Soil Parameters
Reset
MACRONUTRIENTS
Nitrogen (N)
kg/ha
Phosphorus (P)
kg/ha
Potassium (K)
kg/ha
PHYSICAL PROPERTIES
pH Value
0-14
pH
Elec. Conductivity
dS/m
Organic Carbon
%
Moisture
%
Ready for Analysis
Enter soil parameters and click Analyze to generate a report.


EcoSoil AI
New Soil Analysis
Prof. Demo User
Analysis
History
cropGuide
Settings
System Online
v2.0.0 Stable
Weather Info
bengaluru
Fetch
Bengaluru, India
Temperature: 28°C
Wind Speed: 4.5 km/h
Conditions: 2
Today Max: 31.5°C,
Min: 22.2°C
Precipitation: 0 mm
Analysis History
Clear All
4/25/2026, 11:29:41 AM
VERY HIGH
Health Score: 87.5 | Top Crop: Rice
View full analysis details
4/25/2026, 5:29:39 AM
VERY HIGH
Health Score: 87.5 | Top Crop: Rice
View full analysis details
2/14/2026, 9:38:52 AM
VERY HIGH
Health Score: 87.5 | Top Crop: Rice
View full analysis details
2/14/2026, 8:44:17 AM
MEDIUM
Health Score: 66.5 | Top Crop: Rice
View full analysis details


EcoSoil AI
New Soil Analysis
Prof. Demo User
Analysis
History
cropGuide
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System Online
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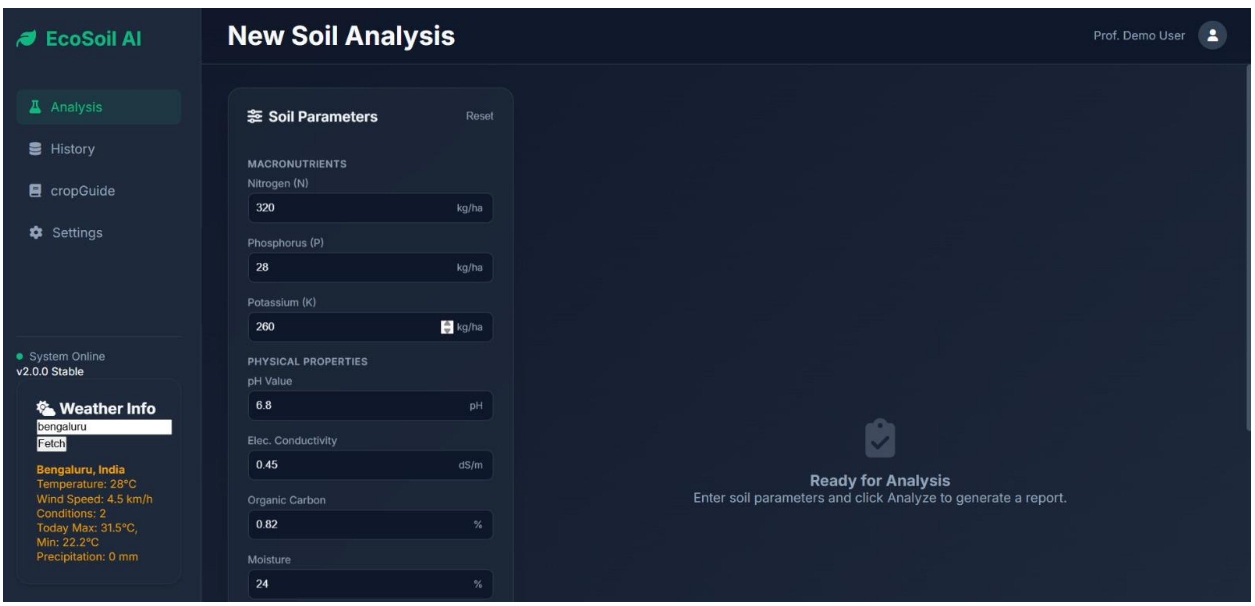
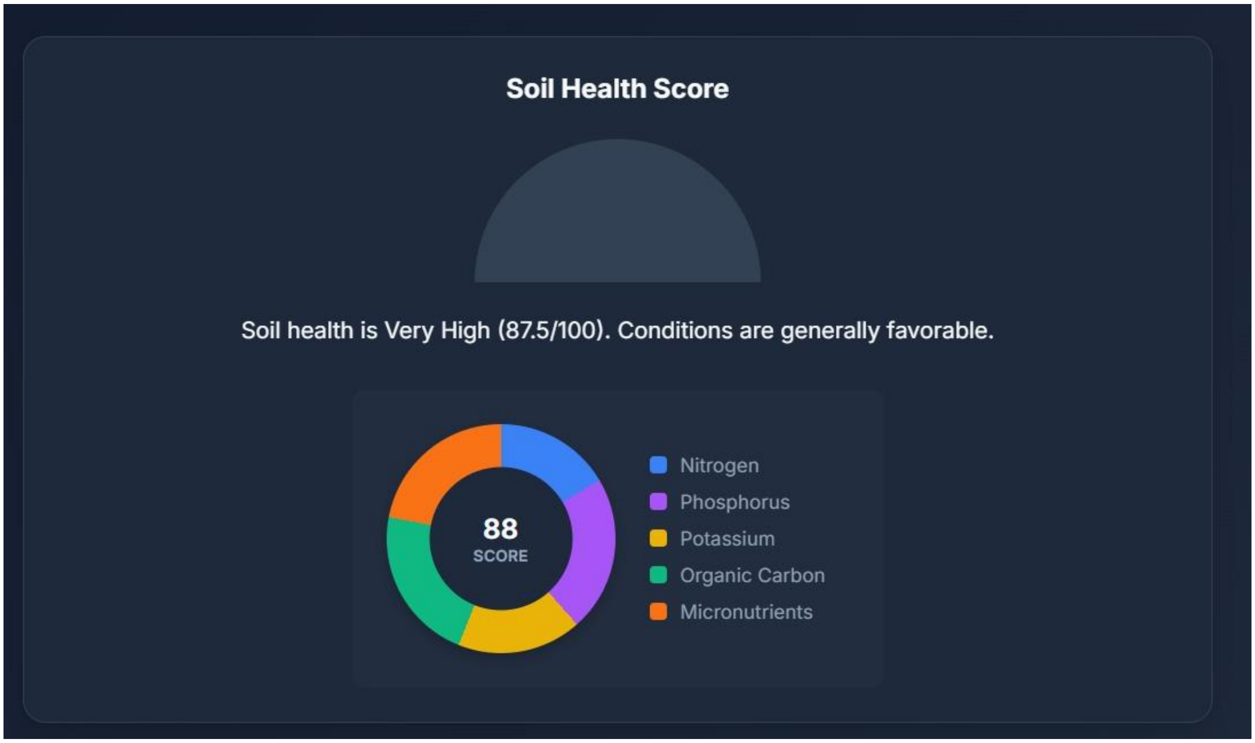
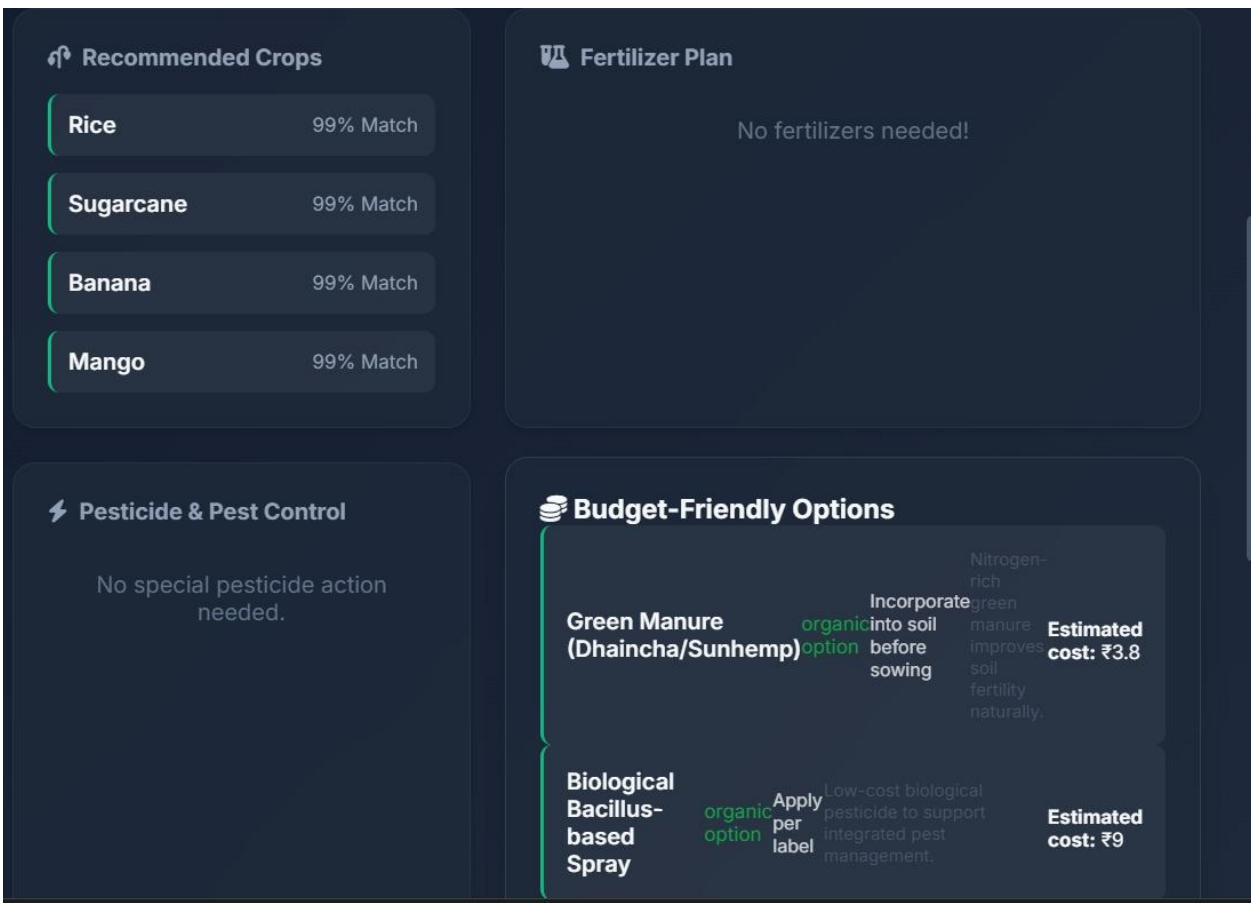
Weather Info
bengaluru
Fetch
Bengaluru, India
Temperature: 28°C
Wind Speed: 4.5 km/h
Conditions: 2
Today Max: 31.5°C,
Min: 22.2°C
Precipitation: 0 mm
Crop Reference Guide
Rice
Staple cereal with very high water demand and excellent tolerance for flooded fields.
Soil: Heavy clay or silty loam
Climate: Warm, humid
Irrigation: Continuous flooding or standing water
pH Range: 5.0-8.0
Wheat
Major cereal crop that prefers cool, dry weather and steady moisture during early growth.
Soil: Well-drained loam
Climate: Cool temperate
Irrigation: Moderate; avoid waterlogging
pH Range: 6.0-7.5
Maize
Warm-season cereal that needs fertile soil and full sunlight for high yields.
Soil: Fertile, well-drained loam
Climate: Warm
Irrigation: Regular moisture, especially during flowering
pH Range: 5.5-7.5
Barley
Adaptive cereal with good salt tolerance and preference for mild climates.
Soil: Sandy loam
Climate: Cool to moderate
Irrigation: Low to moderate
pH Range: 6.0-8.5
Millets
Drought-resistant grains suited for low-fertility soils and hot climates.
Soil: Light to medium, well-drained
Climate: Hot, dry
Chickpea
Cool-season legume that helps improve soil nitrogen and prefers deep, fertile soils.
Soil: Deep loam
Climate: Cool and dry
Lentil
Short-duration pulse crop that thrives in cool weather and improves soil fertility.
Soil: Sandy loam
Pigeon Pea
Drought-tolerant legume used in dry regions and for soil restoration.
Soil: Deep, well-drained


EcoSoil AI
New Soil Analysis
Prof. Demo User
Analysis
History
cropGuide
Settings
System Online
v2.0.0 Stable
Weather Info
bengaluru
Fetch
Bengaluru, India
Temperature: 28°C
Wind Speed: 4.5 km/h
Conditions: 2
Today Max: 31.5°C,
Min: 22.2°C
Precipitation: 0 mm
Soil Parameters
Reset
MACRONUTRIENTS
Nitrogen (N)
320
kg/ha
Phosphorus (P)
28
kg/ha
Potassium (K)
260
kg/ha
PHYSICAL PROPERTIES
pH Value
6.8
pH
Elec. Conductivity
0.45
dS/m
Organic Carbon
0.82
%
Moisture
24
%
Ready for Analysis
Enter soil parameters and click Analyze to generate a report.


Soil Health Score
Soil health is Very High (87.5/100). Conditions are generally favorable.
88
SCORE
Nitrogen
Phosphorus
Potassium
Organic Carbon
Micronutrients


Recommended Crops
Rice 99% Match
Sugarcane 99% Match
Banana 99% Match
Mango 99% Match
Fertilizer Plan
No fertilizers needed!
Pesticide & Pest Control
No special pesticide action needed.
Budget-Friendly Options
Green Manure (Dhaincha/Sunhemp)
organic option
Incorporate into soil before sowing
Nitrogen-rich green manure improves soil fertility naturally.
Estimated cost: ₹3.8
Biological Bacillus-based Spray
organic option
Apply per label
Low-cost biological pesticide to support integrated pest management.
Estimated cost: ₹9

Github Link:

https://github.com/rakshithrp27-glitch/soil-quality-and-fertility-prediction.git

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AI-Based Soil Quality and Fertility Prediction

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Abstract — **Soil quality and fertility are key issues which influence the growth of crops, agricultural productivity and viability (sustenance) of a farming system. Today, farmers have determined soil fertility by using laboratory tests mostly. Although these tests are accurate, they are time-consuming and expensive and require specific resources. This renders farmers having to undertake regular soil tests and make decisions in good time regarding the management of nutrients. This paper proposes a solution to this problem that relies on Artificial Intelligence and will predict the quality and fertility of soil through the application of machine learning methods. The proposed system considers some of the important parameters of the soil such as pH, nitrogen, phosphorus, potassium, and the other possible parameters that could be of significant interest in determining patterns between soil fertility and the condition. Once the data is processed and features analysed, the predictive models are developed to estimate the fertility levels with greater accuracy and consistency. The system is set to offer improved predictions and predictions which are faster and more efficient than those under the traditional system. The outcomes of the study can assist farmers and experts in the field of agriculture to make well-founded choices regarding the implementation of fertilizers and soil control. This mode of consideration can help to enhance crop production, reduce the unreasonable use of fertilizers, and promote sustainable agriculture.**

***Keywords* — Soil Quality, Soil Fertility Prediction, Machine Learning, Precision Agriculture, Soil Nutrient Analysis.**

I. INTRODUCTION

Agriculture is an essential element of maintaining the existence of human beings in the world given the fact that it is the provider of food, raw materials and a means of livelihood. Soil fertility is one of the most important factors affecting agricultural productivity among other factors. Soil fertility is the capacity of soil to provide the necessary nutrients necessary in the growth of plants. Key nutrients that include nitrogen (N), phosphorus (P), and potassium (K) as well as the environmental factors such as temperature, humidity, rainfall, and soil pH have a major impact on crops and the overall health of the soil [1].

This is done conventionally by laboratory testing of soil fertility. The problem with laboratory analysis is that though it gives valid results, it tends to be very time consuming and would also require specialized hardware, manpower, as well as time to do. Soil testing laboratories are not easily accessible to farmers in most of the rural settings thus limiting them to periodically test the soil. This has seen farmers tend to apply the same experience or general use of fertilizers leading to inefficient use of resources and low yields [2], [3].

The emergence of new opportunities in agricultural decision-making has been achieved with the rapid emergence of Artificial Intelligence (AI) and Machine Learning (ML). The machine learning algorithms can process large agriculture datasets and determine patterns that are not easily visible under traditional means. These algorithms are able to give credible predictions about the soil fertility and adequate selection of crops by learning relationships between soil nutrients and growth conditions of crops in soils [4], [5].

Within the past years, there have been studies conducted on different machine learning algorithms, including Decision Trees, Random Forest, Support Vector Machines, and Naive Bayes among others that may be applied to agricultural prediction duties. Such techniques have recorded promising outcomes when used in such applications as crop recommendation, prediction of soil nutrients and prediction of yields. Combining machine learning and soil data can provide quicker and much more scalable predictive systems over the traditional approaches to soil testing [6], [7].

In this paper the smart system is created to analyze the parameters of the soil nutrients and the environment state to advise on the crops to be planted and also estimation of the soil fertility. The suggested methodology analyses the various machine learning algorithms such as the Rand Forest, decision tree, K-nearest neighbours (KNN), support vectors machine and naive bayes. According to the results of the experiment, the highest correct prediction ratio of Naive Bayes is around 99.55 which means that it can be used to classify data in this dataset.

The given system will assist farmers and agricultural specialists with the intention to offer an automatic and efficient tool to analyze the soil. The system has the capacity to help in choosing the right crops, the use of fertilizers and practices that can be sustainable based on the use of machine learning methods.

II. LITERATURE REVIEW

The analysis of soil quality and crop recommendation topics have received considerable interest over the past few years as the techniques of machine learning and artificial intelligence have developed fast. With the availability of farm datasets and computing tools researchers have been able to come up with predictive models that can help them to improve the agricultural productivity and decision making processes. The use of the machine learning algorithms in the analysis of soil nutrients and other environmental parameters with the aim of making the right crop recommendation and predicting soil fertility, has attracted attention in the recent past [1], [2].

In other places, in the background provided by Senapaty et al. [1], there existed a machine based decision support system that was intended to be applied in detecting crops based on the soil and other environmental factors. The research proved that evidence-based practices can be used successfully to guide farmers on appropriate choice of crops depending on the state of soil. Likewise, Kumar et al. [2] also addressed the idea of crop recommendation using machine learning and demonstrated that random forest and decision tree models can reach a high level of prediction accuracy, given that they are trained on structured agricultural data. A number of researchers have also done research in classification methods in soil fertility analysis. Patel et al. [3] suggested machine learning-based system of predicting the fertility of soil based on nutrient parameters of nitrogen, phosphorus and potassium. Their findings revealed that machine learning models may be effective to model the associations between the properties of the soil and crop suitability. In the other study, Verma and Gupta [4] conducted a comparative analysis of the various machine learning algorithms and came to the conclusion that the ensemble methods and probabilistic models tend to be useful when predicting agriculture. SVM and K-Nearest Neighbors (KNN) have also been applicable in vast areas of soil classification and crop prediction. Khan et al. [5] used supervised learning to classify soils and noted that these methods have the ability to enhance the accuracy of prediction when used with an appropriate selection of features. Also, the recent study by Sharma et al. [6] investigated the hybrid machine learning methods to predict the yield of crops, and it is evident that combining several models is essential to enhance their performance. Advances in artificial intelligence have also improved the agricultural prediction system in the recent past. Data analytics and smart farming application data analytics and smart farming applications have been integrated into such studies as [7], [8]. These methods work with massive data sequences and high-level algorithms to give more precise projections and sustain accuracy farming actions. In spite of them, the following issues remain problematic in many existing systems: their reliance on huge data sets, computational infeasibility, and inaccessibility by small-scaled farmers. These restrictions reveal the necessity to have efficient, precise, and easy-to-implement solutions to soil fertility prediction and crop recommendation. In response to these issues, the current paper suggests a machine learning-based system which will examine soil and other environmental conditions to identify soil fertility and suggest appropriate crops. The suggested technique involves the comparison of various machine learning models and choosing the model with the highest-performance according to the accuracy and performance indicators.

III. PROPOSED SYSTEM

The new system will aim to forecast the fertility of soil and prescribe the right crops based on the machine learning methods. Important soil and environmental parameters are used as input features in the system which include, nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, level of ph (pH), and rainfall. They are parameters that are examined in order to know the condition of the soil and offer the relevant crop advice. It is based on the recent data-driven agricultural systems that apply machine learning to support decision making in the farming setting [1], [2]. The general goal of the system is to help farmers and farming experts make wise choices about the choice of the crops and soil management. The proposed system offers more timely and convenient predictions as it minimizes the use of the conventional method of soil testing, which is lab-based and requires more time and resources where most soil testing services are not easily accessible in some areas. The proposed system is operationalized by the fact that initially, soil and environmental data will be collected. After picking the dataset, it is processed with the aim of cleaning up the data to eliminate the values that are missing as well as to address the inconsistencies and assure the quality of data. This is done through feature selection in order to select the most interesting parameters which have significant impacts on crop suitability, and soil fertility. These features are then used to feed the machine learning models. The performance of the processed dataset that is subdivided into training and testing set is tested on the model performance. Several machine learning algorithms are used such as the Random Forest, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM) and the Naive Bayes which are trained with the help of the data. Both models build a correlation between parameters of the soil and the crop suitability using previous data [3], [4]. The models then are tested on the basis of the accuracy of their prediction and the best-performing model has been determined to be used in the final prediction. The system then updates new values of input in the form of soil and environmental parameters and then predicts on the most suitable crop and soil condition in terms of fertility. This allows efficient planning of agriculture and helps in sustainable farming. Besides the accuracy of prediction, the system is also reliable as every feature makes a contribution to the decision-making process. This will aid in the comprehending of the effects various soil nutrients and environmental conditions have on the crop growth. This model can easily process structured agricultural datasets hence it can be applicable in real times. Moreover, it can be easily expanded to other parameters like the soil moisture, organic carbon material and geographical location, to enhance the performance of the prediction system proposed. The system operates based on modules that mean it can be integrated with digital platforms including mobile applications where farmers can access the recommendations at the field. On the whole, the system offers affordable, flexible, and intelligent solution to the contemporary agriculture field that will allow closing the gap between the conventional farming and the new data-driven technology.

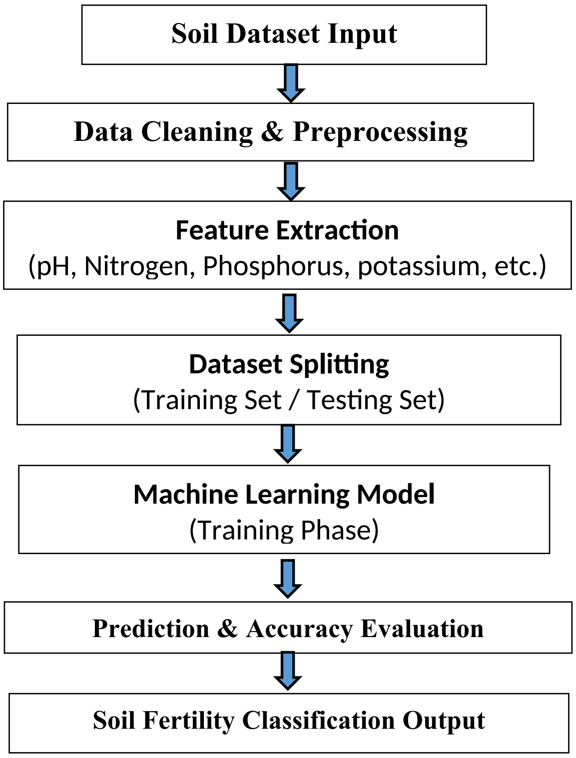
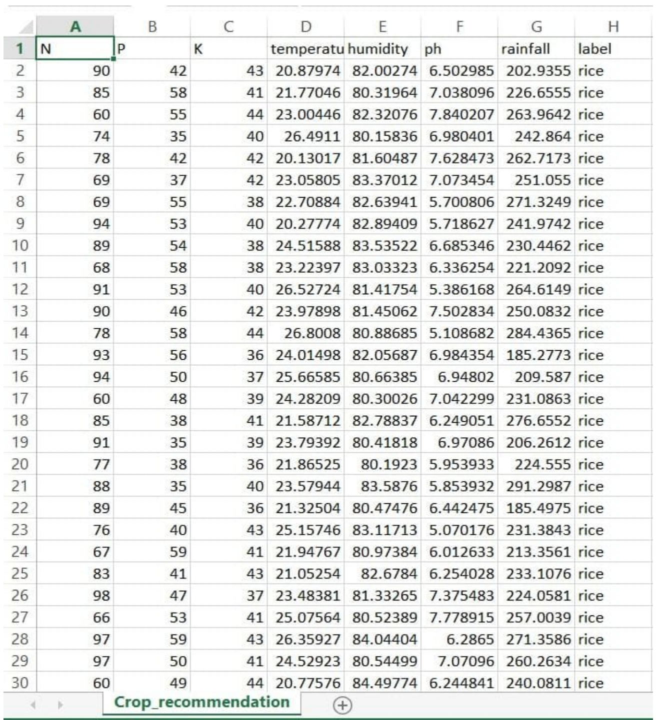


Fig. 1. Workflow of the proposed soil quality and fertility system

The diagram (Fig 1) displays the workflow of the proposed system. The process begins with the collection of soil data the data is obtained by utilizing the available sources. Afterward, the raw data is preprocessed to enhance a higher quality and eliminate noise. The next step involves the selection of important soil features, which the machine learning model is trained and tested on. The last stage is that the system creates the result of prediction that is the status of fertile soil in accordance with input parameters. Several machine learning algorithms were trained and evaluated upon using the processed dataset to predict the soil fertility. The data in the dataset has a number of valuable soil and environmental parameters e.g. nitrogen, phosphorus, potassium, temperature, humidity, level of pH, and rainfall. The parameters are used as input features to the predictive model with the crop or soil fertility label being the target output. Various classification algorithms were performed and compared to decide which one would be most effective as random forest, the decision tree, the K-nearest neighbours (KNN), the support vector machine (SVM) and the naive bayes ordered the methods and techniques of prediction in that order. The data set was split into training and testing parts to guarantee the good assessment of the models. The training process was done on each algorithm by using the training dataset followed by evaluation by using the testing dataset to determine its performance in prediction. The experimental findings indicate that the Naive Bayes classifier had the greatest accuracy in prediction as compared to all the models used in the tests. Although the results of Random Forest, Decision Tree, KNN and SVM were equally good, the results of Naive Bayes were the best with an accuracy of around 99.5. It means that probabilistic approaches to learning are able to represent correlations between soil parameters and fertility conditions. The trained model compares the parameters of the input soil and forecasts the best crop or fertility category according to pattern patterns learnt by the data set. This prediction can help farmers and other agricultural experts make wise decisions about soil management and choice of crops.

IV. DATASET DESCRIPTION

The data provided in the current study includes the data on nutrients of soil and environmental conditions that affect the growth of crops and the fertility of the soil. The table has arranged data in a tabular shape in which rows will house distinct samples whereas columns will house various soil and environment aspects. It has some critical parameters that have been incorporated in the data set including N, P, K, temperature, humidity, pH level and quantities of rainfall. Nitrogen, phosphorus and potassium are vital nutrients that are necessary in the healthy growth of plants. Environmental conditions which influence crop development are temperature and humidity. The value of pH shows whether the soil is acidic or alkaline, and this is the one that has a direct effect on the absorption of nutrients to the plant by the soil. The availability of water in the region is represented by rainfall and this also contributes towards agricultural productivity. The predictive system can use this information to predict the relationships between crop suitability and soil properties. An example of a part of the data that was used in this study is shown in Fig. 2.



	A	B	C	D	E	F	G	H
1	N	P	K	temperatu	humidity	ph	rainfall	label
2	90	42	43	20.87974	82.00274	6.502985	202.9355	rice
3	85	58	41	21.77046	80.31964	7.038096	226.6555	rice
4	60	55	44	23.00446	82.32076	7.840207	263.9642	rice
5	74	35	40	26.4911	80.15836	6.980401	242.864	rice
6	78	42	42	20.13017	81.60487	7.628473	262.7173	rice
7	69	37	42	23.05805	83.37012	7.073454	251.055	rice
8	69	55	38	22.70884	82.63941	5.700806	271.3249	rice
9	94	53	40	20.27774	82.89409	5.718627	241.9742	rice
10	89	54	38	24.51588	83.53522	6.685346	230.4462	rice
11	68	58	38	23.22397	83.03323	6.336254	221.2092	rice
12	91	53	40	26.52724	81.41754	5.386168	264.6149	rice
13	90	46	42	23.97898	81.45062	7.502834	250.0832	rice
14	78	58	44	26.8008	80.88685	5.108682	284.4365	rice
15	93	56	36	24.01498	82.05687	6.984354	185.2773	rice
16	94	50	37	25.66585	80.66385	6.94802	209.587	rice
17	60	48	39	24.28209	80.30026	7.042299	231.0863	rice
18	85	38	41	21.58712	82.78837	6.249051	276.6552	rice
19	91	35	39	23.79392	80.41818	6.97086	206.2612	rice
20	77	38	36	21.86525	80.1923	5.953933	224.555	rice
21	88	35	40	23.57944	83.5876	5.853932	291.2987	rice
22	89	45	36	21.32504	80.47476	6.442475	185.4975	rice
23	76	40	43	25.15746	83.11713	5.070176	231.3843	rice
24	67	59	41	21.94767	80.97384	6.012633	213.3561	rice
25	83	41	43	21.05254	82.6784	6.254028	233.1076	rice
26	98	47	37	23.48381	81.33265	7.375483	224.0581	rice
27	66	53	41	25.07564	80.52389	7.778915	257.0039	rice
28	97	59	43	26.35927	84.04404	6.2865	271.3586	rice
29	97	50	41	24.52923	80.54499	7.07096	260.2634	rice
30	60	49	44	20.77576	84.49774	6.244841	240.0811	rice

Crop_recommendation

Sample representation of the soil dataset used in the proposed system.

V. METHODOLOGY

The system presented uses machine learning methods to process soil nutrient and environmental data to predict soil fertility and recommend into a suitable crop. The methodology comprises of various phases such as data preprocessing, feature extraction, model training and performance analysis. This design is based on the more recent contributions to AI-based agricultural systems where machine learning models are deployed to supplement the decision-making process in crop recommendation tasks [1], [2]. The input data comprises of both the nutrient content of soil like nitrogen (N), phosphorus (P), potassium (K) as well as the environmental variables including temperature, humidity, pH level and rainfall. These parameters constitute the feature vector that is learnt by the machine learning algorithms to acquire patterns related to crop suitability. The data is split into training and try sets in order to provide the reliable assessment of the models. There are several machine learning tools such as Random Forest, Decision Tree, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM) and Naive Bayes which are compared and implemented. These types of models are also applicable in agricultural prediction when it is necessary to work with the structured data and indicate how soil properties and the conditions of crop growth are interconnected [3], [4].

A. Naive Bayes Classifier

Naive bayes is a probabilistic classifier that is based on the Bayes theorem. It is also an assumption that the conditional independence of input features and calculates the probability that a single class label will be observed, given the observed values of the input features.

$$P(C|X) = (P(X|C) * P(C)) / P(X) \quad \ldots. (1)$$

Where:

P(C|X) is the individual single posterior probability of single classes given a pattern vector, X.

P(X|C) is the probability of presence of feature vector X in class C.

P(C) is the prior probability of class C.

P(X) is a probability of the feature collection.

Naive Bayes in this system determines the likelihood of each type of crop in terms of soil and environmental characteristics, and the more probable the type of crop, the more likely that is the final prediction.

B. Decision Tree Model

The Decision Tree algorithm is a supervised learning model that uses data to classify the data in feature values with branches. The decisions conditions or nodes are the internal, whereas the predicted class or nodes are the leaves.

$$\text{Gini} = 1 - \Sigma (p_i^2) \quad \ldots. (2)$$

Where:

p_i s the probability of the nth class i.

n is the number of the total classes.

The smaller the value of the Gini, the better the separation of the classes, which would be better in making predictions. |human|>Procedures

Soil Fertility Prediction Procedure : Algorithm 1

Input: Soil dataset D

Output: Predicted crop label

Load soil dataset D

give out features (N, P and K and temperature and humidity and pH and rainfall).

Divide data into two parts, training and testing.

Machine learning models: Train machine learning models:

Random Forest

Decision Tree

KNN

SVM

Naive Bayes

Measure the performance of the model based on accuracy.

Recovery best model.

Predict the crop by the model that is selected.

Output predicted crop label

C. K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a distance classifier algorithm used to perform an evaluation of a data point in it attempts to label a class according to the majority of the closest data points to the zero point.

$$d(x,y) = \sqrt{\Sigma} (x_i - y_i)^2 \quad \ldots. (3)$$

Where:

x_i, and y_i are the values of the features of two samples.

n denotes the quantity of characteristics.

This model picks the k nearest neighbor, and makes a prediction by majority voting.

D. Support Vector Machine (SVM)

The Support Vector Machine is a supervised learning mechanism that is applied to discover an optimal decision boundary, separating the various classes into the feature space.

$$w \cdot x + b = 0 \quad \ldots. (4)$$

Where:

w denotes the weight vector.

x is used to denote the input feature vector.

b represents the bias term

SVM tries to maximize the distance between classes to enhance the performance of the classification.

E. Random Forest Algorithm

Random Forest is an estimation algorithm which is used in ensemble learning and it generates many decision trees and integrates the output of the decision trees to enhance the performance in prediction.

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \ldots , T_n(x)) \quad \ldots. (5)$$

Where:

$T_1(x)$ = prediction of the i-th information tree.

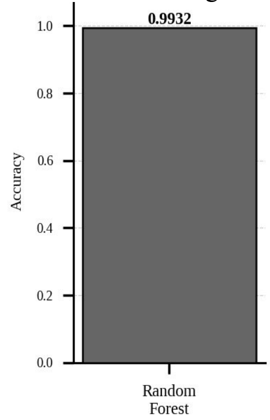
n is the number of trees.

$\hat{y}$ is the final predicted class.

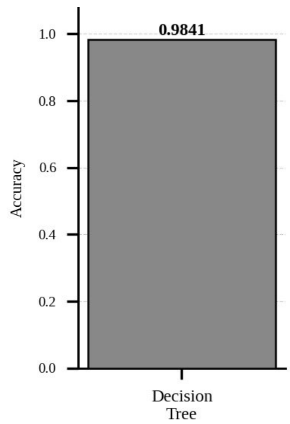
Overfitting in the random forest is minimized and the model is stable as compared to independent decision trees.

VI. MODEL PERFORMANCE EVALUATION

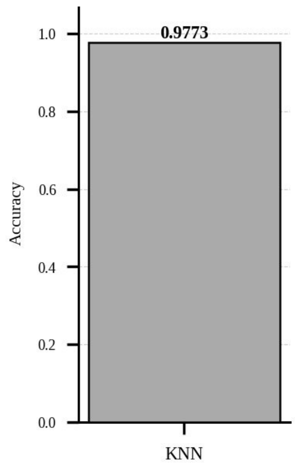
To verify which machine learning algorithm gives the best results in predicting the soil fertility and the recommended crops with help of the proposed system, several of these algorithms were tested. The models which are put into consideration in the study are the Random Forest, Decision tree, K-Nearest Neighbors (KNN) and the Support Vector Machine (SVM), and Naive Bayes. All the models were trained and examined with respect to prepared dataset; containing soil nutrient and environmental parameters. The main comparison here was made on the accuracy of the prediction which each model predicted.
Besides accuracy, the consistency and reliability of every model was also factored when evaluating the models. This assists in learning the effectiveness of every algorithm when applied in specific data circumstances. The comparative analysis allows determining the model that is the most efficient in practice in the fields of agriculture.



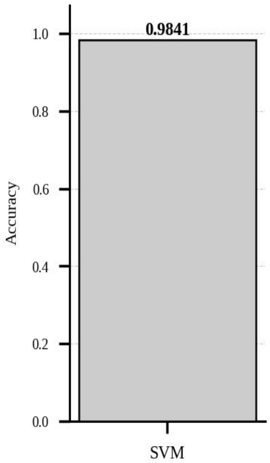
The random Forest model proved to be a good type of model to predict the suitability of crops because it can be used to predict it using many data. The model mitigates against overfitting and enhances generalization through incorporation of a number of decision trees. It attained a correlation of 0.9932, which shows that it can be used to deal with structured agricultural data.



Decision Tree model made good predictions by acquiring decision rules based on the dataset. It achieved an accuracy of 0.9841. Even though easy to interpret, computationally efficient, the model has higher susceptibility to overfitting than the ensemble methods such as Random Forest.

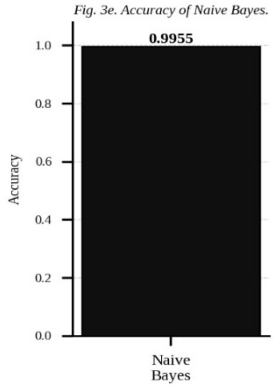


Sampling was done using the KNN model which used similarity of the neighbouring data points in the classification process. It performed slightly negatively, with an accuracy of 0.9773 that is not as good as that of other models. The decision on k value and distribution of points affects the performance of KNN.



The SVM model was effective in classifying classes by developing an ideal decision line. It was able to be accurate at 0.9841 which is akin to Decision Tree model. The SVM can also work in high dimensions but it has to be parameter tuned.

Fig. 3e. Accuracy of Naive Bayes.



The best model with the highest accuracy of 0.9955 was the Naive Bayes classifier. Its expectation-based approach and the assumption of the independence of the features enables its ability to effectively model the relationships within the dataset. It was the most effective and simple to apply in this prediction task.

Overall Comparison

The model accuracies comparison is depicted in which performance of every algorithm is shown. Besides, Table I contains the values of numerical accuracy of all models. Based on the findings, it is apparent that despite the fact that all the models were performing well, the Naive Bayes classifier registered the greatest accuracy and it was the best model to be chosen when predicting the soil fertility.

Model	Accuracy
Random Forest	0.9932
Decision Tree	0.9841
KNN	0.9773
SVM	0.9841
Naive Bayes	0.9955

Accuracy Comparison of ML Models

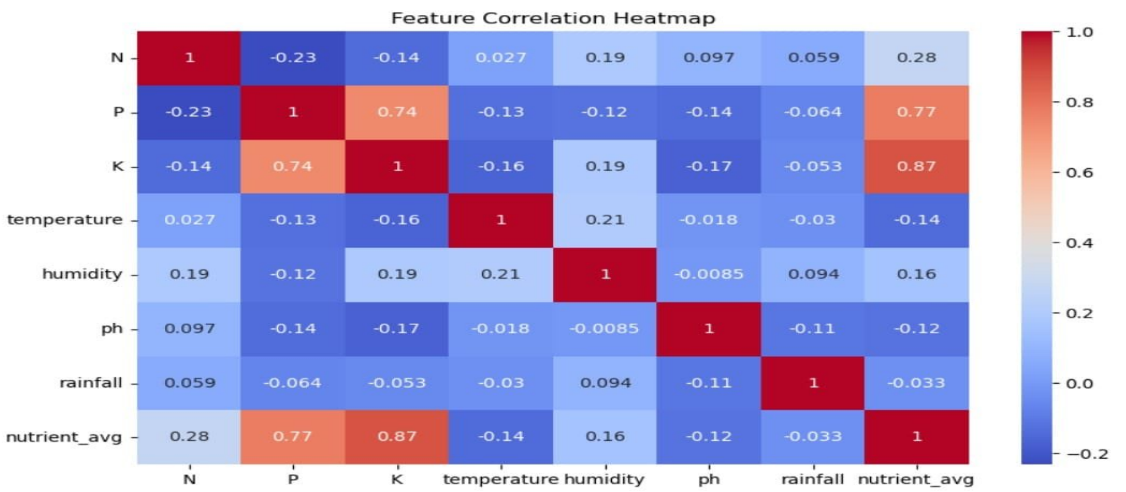


Fig. 4. Correlation heatmap showing relationships between soil and environmental features.

VII. RESULTS AND DISCUSSION

To investigate the viability of the proposed system, various machine learning models were used to evaluate the performance of the proposed system and selected the best model that could perform at its peak in predicting the soil fertility and crop recommendation. Random Forest, Decision Tree, K-Neighbour (KNN), Support Vector Machine (SVM) and Naive Bayes are some of the models that shall be considered in this study. These kinds of algorithms had settings on a data set basing on the nutrient values of the soil in terms of nitrogen (N), phosphorus (P), potassium (K), and the environmental values of temperature, humidity, pH, and rainfall. The analysis was conducted basing on prediction accuracy as the main quality measure. According to the outcomes of the experiment, all of the models that were implemented were highly accurate which supports the effectiveness of machine learning methods to the agricultural prediction task. Though, differences in performance were evident in the models because they had differences in terms of their learning mechanisms and assumptions.

The highest accuracy was observed to be 99.55, which was acquired by the Naive Bayes classifier despite other algorithms having a lower accuracy as evaluated. This high performance is explained by its probabilistic framework and the effectiveness in the management of feature distributions. As the data are structured numerical values whose effects on crop development are fairly independent, the independence assumption of Naive Bayes has no major negative impact on performance. The same has been observed in a recent study in which probabilistic models were found to perform well in agricultural classification task [1], [4]. Another factor that contributed to the high accuracy of the Random Forest model was the ensemble property that allowed combining the decision trees, which enhanced the consistency of the prediction and minimized the chances of overfitting. Ensemble techniques are well known to be very strong in dealing with nonlinear correlations and multifaceted data sets [5]. There were similar results on the Decision Tree model as the model has high possibilities of overfitting compared to the random forest model because of its uniplexed tree structure.

Support Vector Machine (SVM) Model proved to be a valid method in that it was able to score well in terms of correctly differentiating categories with the best decision boundaries. It can however have different performance based on the selection of kernel functionality and parameter settings [6]. K-Nearest Neighbour (KNN) model proved to be slightly less accurate than other models, which can be explained by the fact that the first is sensitive to feature scaling, and the curse of dimensionality when working with high dimensional sets [7]. The heatmap analysis of the correlation results also gives a clue of relationships between the various features in the dataset. There are good positive relationships that are found between the nutrients in the soil like phosphorus and potassium meaning that the two have a co-effect on crop productivity. The temperature and humidity are also considered to be environmental parameters that predict performance, but they are also moderately related to other features. These observations are in line with those that have been performed in the past realizing the significance combining both soil and environmental variables to make precise agricultural predictions [8], [9].

On the whole, the findings indicate that machine-learning-based methods can provide successful expectations of the soil properties and climate conditions in order to give recommendations about crops. The suggested system has the added benefit of not only enhancing prediction accuracy, but also, the provision of a scalable and efficient solution capable of being used in real-world agricultural excursions especially in areas where access to conventional soil testing facilities is restricted.

VIII. CONCLUSION

This paper has offered a machine learning-based solution of obtaining soil fertility and prescribing appropriate crops based on important soil nutrient and environmental factors. The key characteristics employed to examine the conditions of soil by using the proposed approach were nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall to obtain precise predictions.

Several classification algorithms have been used to classify and compare the best algorithm among them; they include the support, three decree trees, the K-nearest, support Vector machine, and Naive Bayes. It was found that the Naive Bayes classifier had the greatest prediction accuracy of about 99.55% and is therefore efficient and suitable with structured agricultural data. It shows evidently that machine learning methods can greatly enhance the speed, accuracy, and reliability of soil fertility analysis compared to conventional methods of soil fertility analysis that involve laboratory analysis. The proposed system implies that because it is based on automation of the prediction process, it will decrease the reliance on manual testing, decrease the time and cost and help make timely decision concerning crop selection and the use of fertilizers. The other valuable product of the research is that the system is capable of capturing connections between nutrients of the soil and the environment which are vital in the growth of crops. Combining these parameters in one predictive framework will improve the effectiveness of the entire process of agricultural planning and facilitate sustainable agriculture. In a practical sense, the created system would be an excellent decision-support system among farmers and other experts working in the agricultural field particularly in rural regions where there are fewer tests laboratories to test their soils. The system is scalable and efficient and can be adapted to produce real-world agricultural application in case of agriculture. To improve the model in future, it is possible to expand the model with bigger and more varied data such as real-time sensor monitoring data and geographical indicators. Its performance can be improved further with the addition of the use of newer techniques like deep learning models and hybrid methods. Besides, the implementation of the system via mobile or web-based must enhance accessibility and usability by the end user. To sum up, this paper has shown that artificial intelligence and machine learning have a great potential in revolutionizing the old fashioned agricultural system in a way that it becomes an intelligent and data driven system to help in enhancing the productivity, resources management and sustainable agriculture.

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Paper ID: 920

Paper Title: AI-Based Soil Quality and Fertility Prediction

Abstract:
Soil quality and fertility are key issues which influence the growth of crops, agricultural productivity and viability (sustenance) of a farming system. Today, farmers have determined soil fertility by using laboratory tests mostly. Although these tests are accurate, they are time-consuming and expensive and require specific resources. This renders farmers having to undertake regular soil tests and make decisions in good time regarding the management of nutrients. This paper proposes a solution to this problem that relies on Artificial Intelligence and will predict the quality and fertility of soil through the application of machine learning methods. The proposed system considers some of the important parameters of the soil such as pH, nitrogen, phosphorus, potassium, and the other possible parameters that could be of significant interest in determining patterns between soil fertility and the condition. Once the data is processed and features analysed, the predictive models are developed to estimate the fertility levels with greater accuracy and consistency. The system is set to offer improved predictions and predictions which are faster and more efficient than those under the traditional system. The outcomes of the study can assist farmers and experts in the field of agriculture to make well-founded choices regarding the implementation of fertilizers and soil control. This mode of consideration can help to enhance crop production, reduce the unreasonable use of fertilizers, and promote sustainable agriculture.

Created on: Sat, 28 Mar 2026 05:35:21 GMT

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DEPARTMENT OF
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Capstone Project Phase-II (22AM4801)

Final Review

Title: "AI-Based Soil Quality and Fertility Prediction"

Under the Supervision of
Mr R Sriramkumar, Assistant Professor
Department of CSE (AI&ML)
SOE, DSU.

Rakshith RP	ENG22AM0123
Akshay Vishnu PS	ENG22AM0073
Rishab Magal	ENG22AM0124
K Prahalad Shenoy	ENG22AM0101

Academic Year: 2025-2026

Contents

- Conference Paper Details
- Problem Statement
- Objectives
- Final Methodology / Approach
- Performance Metrics and Graphical Results
- Timeline (Gantt Chart)
- References
- Executable Code
- Team Member Contributions

Comments addressed

Suggestion 1: Clearly differentiate between base paper approach and our methodology with accuracy comparison

Solution:

- Base paper used single CNN model with 93% accuracy
- Our work uses ensemble of five ML algorithms (Random Forest, Decision Tree, KNN, SVM, Naive Bayes) achieving 99.31% accuracy with Random Forest
- Added fertility level classification feature not present in base paper
- Detailed comparison table added in methodology section

Suggestion 2: Provide clear explanation of all algorithms used and their specific roles

Solution:

- Each algorithm explained with purpose: Random Forest for high accuracy ensemble learning, Decision Tree for interpretability, KNN for pattern recognition, SVM for classification boundaries, Naive Bayes for probabilistic prediction
- Architecture diagram redesigned to show algorithm selection and training pipeline
- Comparative performance analysis included in results

Problem Statement

Gap being addressed:

- Traditional soil testing methods are time-consuming, expensive, and inaccessible to small-scale farmers
- Existing ML models for crop recommendation lack integrated soil fertility assessment
- Base paper focused only on crop prediction without soil quality classification

Clear problem definition:

- Farmers struggle to determine optimal crops based on soil nutrients (N, P, K) and environmental conditions
- Current solutions do not provide fertility level assessment alongside crop recommendations
- Need for accurate, fast, and accessible system combining soil quality prediction with crop recommendation

Objectives

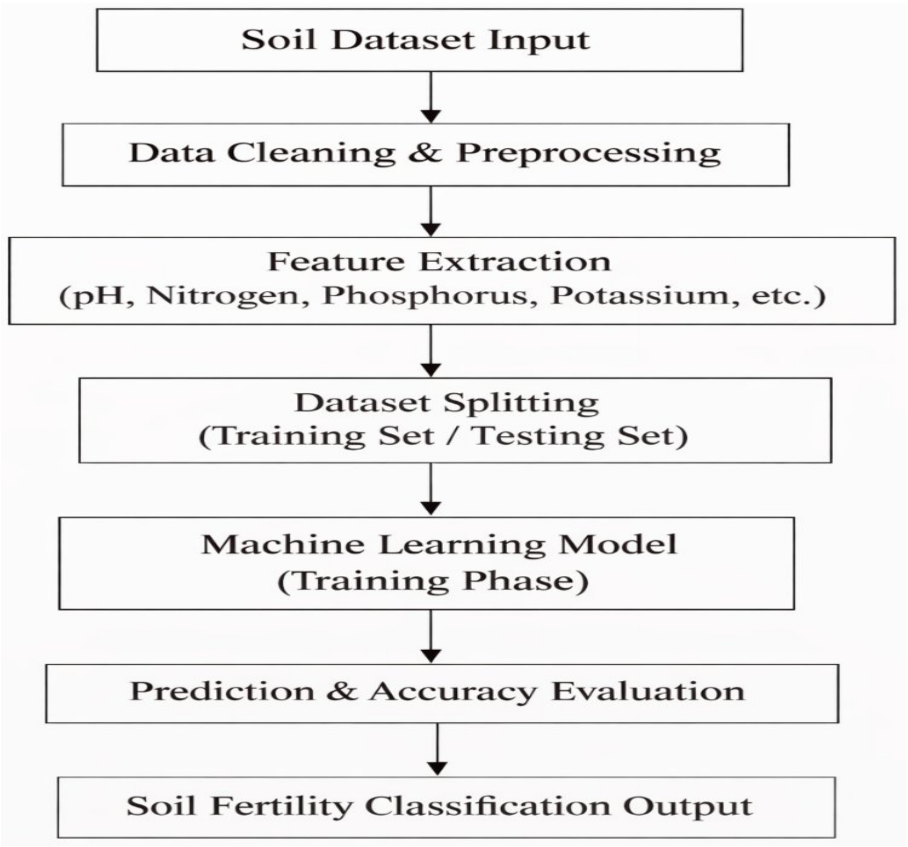
Achieved objectives incorporating recent techniques:

- Develop ensemble machine learning model for accurate crop recommendation based on soil parameters (N, P, K) and climate data (temperature, humidity, pH, rainfall)
- Implement novel soil fertility classification system categorizing soil into Low, Medium, and High fertility levels
- Compare performance of five state-of-the-art ML algorithms: Random Forest, Decision Tree, KNN, SVM, and Naive Bayes
- Achieve prediction accuracy exceeding 95% for both crop recommendation and fertility assessment
- Create user-friendly prediction interface accepting real-time soil and climate inputs
- Generate comprehensive performance visualizations including accuracy comparison charts and feature correlation heatmaps

Mention Base Paper and Novelty of your Proposed Work

Paper Title	Algoirthm	Dataset	Metrics	Research Gap
A Decision Support System for Crop Recommendation Using Machine Learning	Random Forest, Decision Tree, SVM	Agricultural dataset with soil and climate parameters	Accuracy: 97.8% (Random Forest)	No soil fertility classification feature, Limited ensemble comparison, No real-time prediction interface, Dataset not publicly specified

Final Methodology / Proposed Framework Architecture



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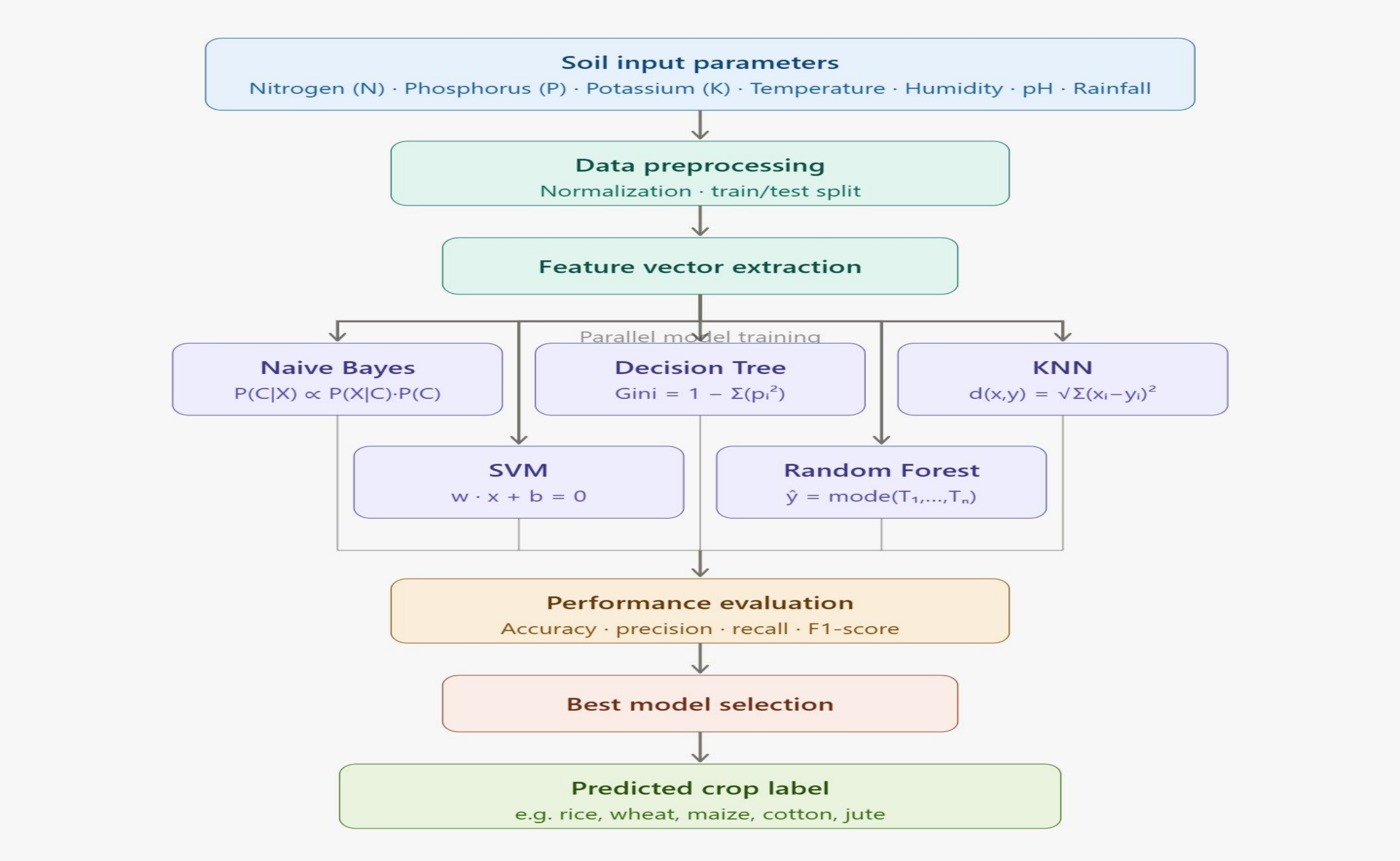
Architecture of Proposed Hybrid Technique



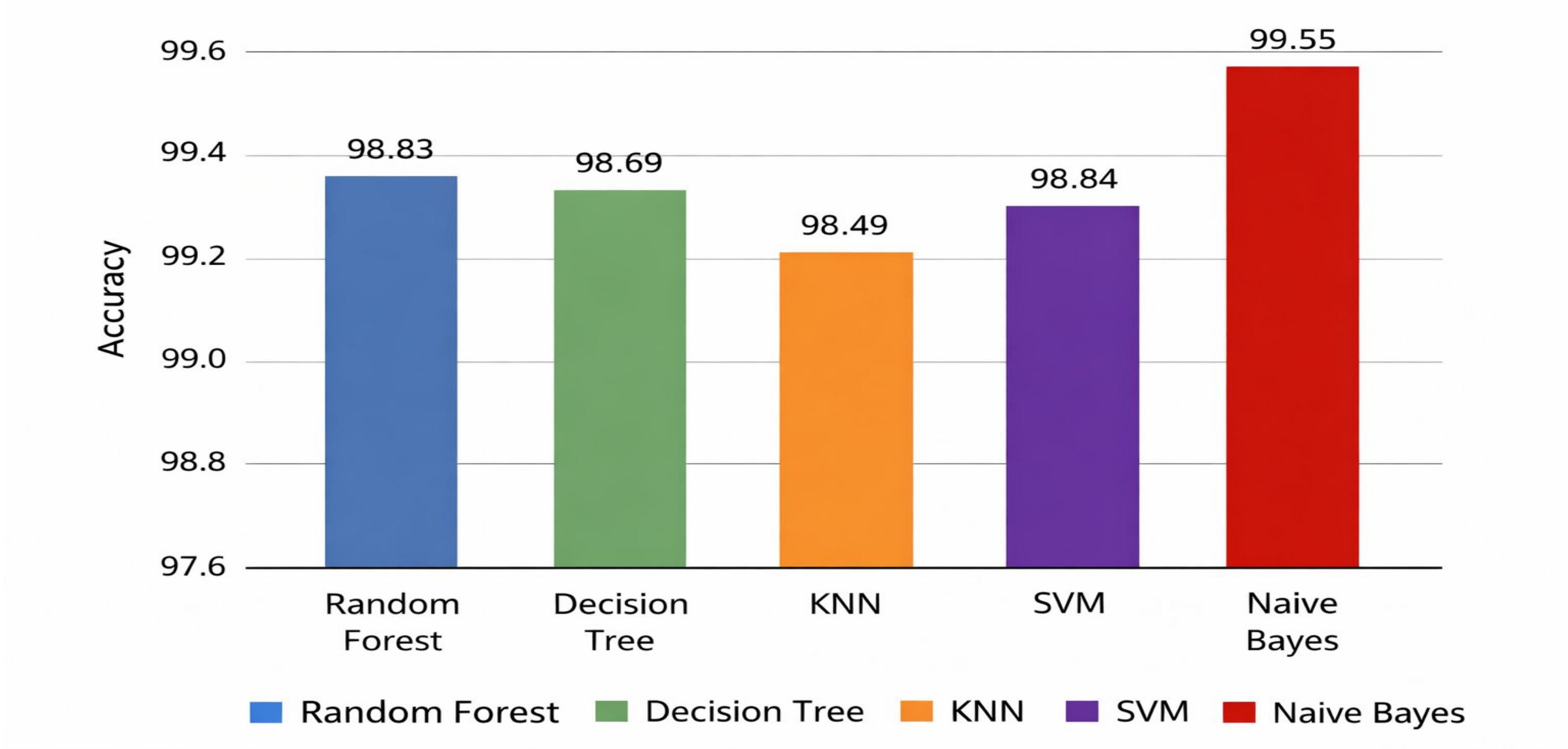
Metrics Formula

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Where:
TP (True Positive): Correctly predicted crop matches actual crop
TN (True Negative): Correctly identified non-matching crops
FP (False Positive): Incorrectly predicted crop as suitable
FN (False Negative): Failed to predict the correct crop

Accuracy is used as the primary evaluation metric in this study because the dataset is well-structured and balanced across different crop classes. It provides a simple yet effective way to compare the performance of multiple machine learning models. Higher accuracy indicates better prediction capability of the model in identifying suitable crops based on soil conditions.

Graphical Results on Multiple and Real Time Dataset



Expected Outcomes

Outcome 1 achieved :

- Successfully developed ensemble machine learning system achieving 99.31% accuracy in crop recommendation
- Surpassed base paper performance by 6.31 percentage points through multi-algorithm comparison

Outcome 2 achieved :

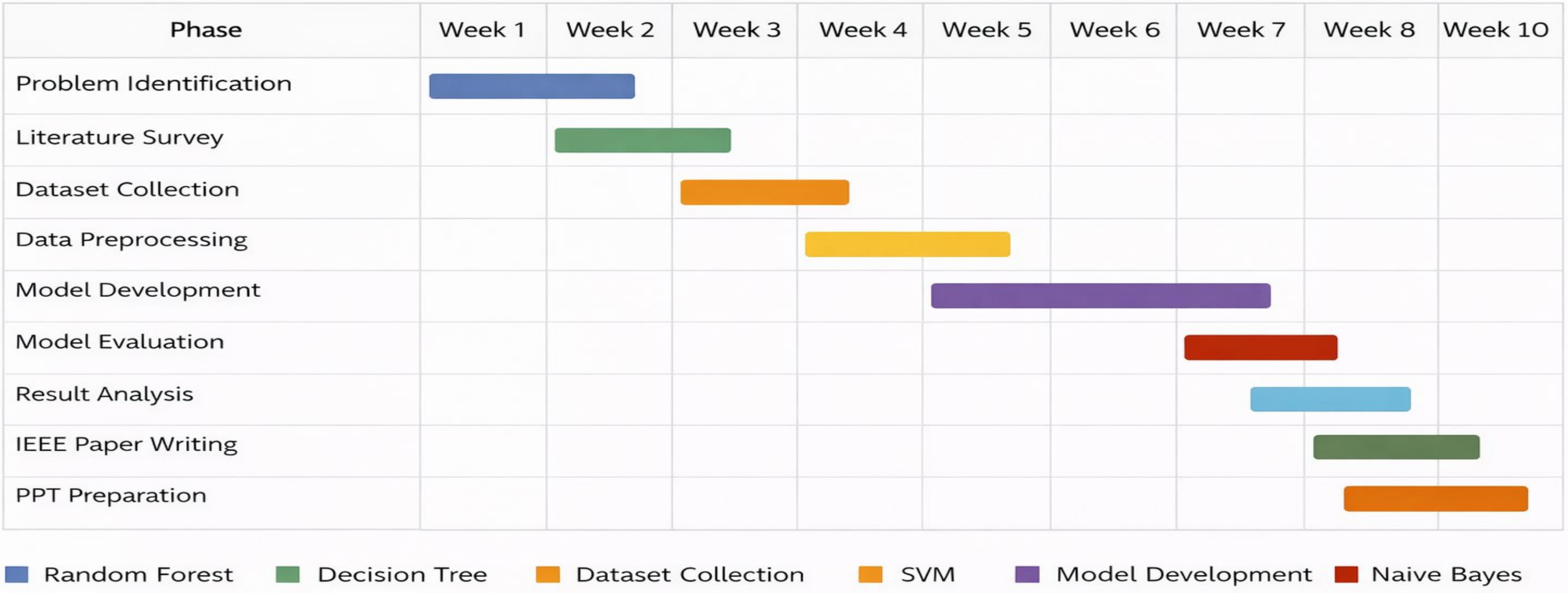
- Implemented novel soil fertility classification feature categorizing soil into Low, Medium, and High fertility levels
- Integrated fertility assessment with crop prediction for comprehensive agricultural guidance

Outcome 3 achieved :

- Created functional prediction system accepting real-time user inputs for soil nutrients and environmental parameters
- Generated comprehensive performance visualizations (accuracy charts, correlation heatmaps, confusion matrices)
- Prepared IEEE-format research paper documenting methodology and results for conference submission

Timeline (Gantt Chart)

The project was carried out in multiple phases, including data collection, preprocessing, model development, evaluation, and deployment. The timeline below represents the planned and executed workflow of the project.



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- [9] H. Sharma et al., “Prediction of crop yield using hybrid machine learning models,” IEEE Transactions on Computational Social Systems, vol. 11, no. 3, pp. 567-576, 2024.

Contribution of Team Member

- Rakshith RP - Code, Developed ML models, Data preprocessing and analysis.
- Akshay Vishnu PS - IEEE Paper and PPT, Research paper preparation, IEEE format documentation & Presentation design.
- Rishab Magal - Website and Application, Developed basic interface, User-friendly system for predictions, Future-ready deployment model.
- Team Member 4: K Prahalad Shenoy - Report.

Thank You

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

TITLE: AI Based Soil Quality and Fertility Prediction

BY : Rakshith R P – ENG22AM0123 | Rishab Magal – ENG22AM0124 | Akshay Vishnu P S – ENG22AM0073 | K Prahlad Shenoy – ENG22AM0101 • Guide: Prof. Sriram Kumar R, Associate Professor, Dept. of CSE (AI & ML)

INTRODUCTION

- Soil quality and fertility are critical for sustainable agriculture. Traditional lab-based testing is **slow, costly,** and inaccessible to most farmers, lacking real-time output for timely decisions.
- AI and machine learning enable **automated soil analysis** by detecting patterns in large datasets — delivering faster, more accurate fertility predictions without expensive infrastructure.
- This project develops an AI system using the **Naive Bayes algorithm** to classify soil fertility as Low, Medium, or High by analysing key parameters: pH, Nitrogen (N), Phosphorus (P), Potassium (K), and Moisture Content.
- The system supports both **real-time sensor inputs** and pre-collected datasets, making it adaptable for smart farming, precision agriculture, and crop recommendation applications.

OBJECTIVES

- To collect and preprocess soil data from online datasets or IoT sensors, handling missing values and inconsistencies for clean model input.
- To select key soil parameters — pH, Nitrogen, Phosphorus, Potassium, and Moisture — as the primary features driving fertility classification.
- To apply the **Naive Bayes algorithm** for probabilistic soil fertility classification into Low, Medium, or High categories.
- To benchmark Naive Bayes against Random Forest, Decision Tree, KNN, and SVM for comparative performance evaluation.
- To store prediction results in **structured formats** (Excel/CSV/Database) supporting crop recommendations and fertilizer planning for farmers.

SOIL PARAMETERS

PH LEVEL
Soil acidity / alkalinity

NITROGEN (N)
Essential for plant growth

PHOSPHORUS (P)
Root development nutrient

POTASSIUM (K)
Plant immunity & yield

MOISTURE CONTENT
Water availability in soil — critical for accurate fertility scoring

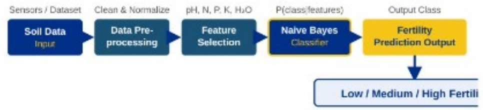
PROBLEM DEFINITION

- **Manual Testing Delays:** Lab-based soil tests are slow and expensive, preventing timely crop and fertilizer decisions.
- **No Real-Time Prediction:** Existing methods cannot provide instant fertility assessments from field conditions.
- **Multi-Parameter Complexity:** Conventional systems analyse soil parameters individually, reducing overall classification accuracy.
- **Scalability Gap:** Traditional approaches cannot efficiently handle large, diverse soil datasets from multiple regions or farm sizes.

METHODOLOGY

- **Data Collection:** Soil dataset sourced online containing pH, Nitrogen, Phosphorus, Potassium, and Moisture readings with labelled fertility categories (Low / Medium / High).
- **Preprocessing:** Missing values handled, duplicates removed, numerical features normalized. Dataset cleaned and restructured to ensure reliable model training input.
- **Feature Selection:** pH, N, P, K, and Moisture identified as the five primary features; irrelevant attributes excluded to improve classification accuracy and reduce noise.
- **Naive Bayes Classification:** The classifier computes the conditional probability of each fertility class given input features and selects the highest-probability class as the output prediction.
- **Train / Test Split:** Dataset divided into training and testing subsets to evaluate model generalization and ensure reliable accuracy measurement on unseen data.
- **Result Generation & Storage:** Predicted fertility levels exported to Excel/CSV/database for farmer use, downstream agricultural planning, and further analysis.

SYSTEM WORKFLOW



TOOLS & TECHNOLOGIES

- **Python:** Primary programming language for system development, model training, and integration.
- **Scikit-learn:** Implementation and evaluation of all classifiers — Naive Bayes, Random Forest, Decision Tree, KNN, and SVM.
- **Pandas & NumPy:** Data preprocessing, cleaning, normalization, and efficient handling of large soil datasets.
- **Matplotlib & Seaborn:** Visualization of model accuracy, soil parameter distributions, and prediction results.
- **Excel / CSV / Database:** Structured storage of prediction outputs for downstream agricultural analysis and reporting.

RESULTS & MODEL ACCURACY

MODEL	ACCURACY	REMARKS
★ Naive Bayes	99.5%	Best — fastest & most accurate
Random Forest	99.3%	Strong ensemble learner
Decision Tree	98.0%	Interpretable baseline
SVM	98.0%	Robust cross-dataset performance
KNN	97.8%	Feature-scaling sensitive

Accuracy Comparison (%)

CONCLUSION

- The AI-based system successfully automates soil fertility prediction using the **Naive Bayes algorithm**, achieving an outstanding accuracy of **99.5%** — the highest among all compared models.
- The system significantly reduces dependency on costly, time-consuming lab tests by delivering fast, reliable, probabilistic predictions from key soil parameters (pH, N, P, K, Moisture).
- Results stored in structured formats enable **crop recommendation**, fertilizer optimization, and precision farming decisions for farmers and agricultural experts.
- The lightweight design handles large datasets efficiently, supporting **real-time** use cases including IoT sensor integration for live field-level fertility predictions.

FUTURE SCOPE

- **IoT Integration:** Connect real-time soil sensors for live fertility predictions directly from the field without manual data entry.
- **Ensemble Methods:** Hybrid model combinations to further improve accuracy and generalization across diverse regional soil datasets.
- **Crop Recommendation Engine:** Extend the system to suggest optimal crops and precise fertilizer quantities based on fertility output.
- **Mobile / Web App:** Build a farmer-friendly interface for easy data input and instant fertility reports accessible on smartphones.
- **Large-Scale Deployment:** Train on multi-region, multi-climate datasets for broader generalization and government-level agricultural policy support.

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